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Topical Review

Optimizing electrical output performance of triboelectric nanogenerators by micro-/nano-morphology design and fabrication

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Abstract

Fueled by the increasing imperative for sustainable energy solutions and the burgeoning emphasis on health awareness, self-powered techniques have undergone notable strides in advancement. Triboelectric nanogenerators (TENGs) stand out as a prominent device capitalizing on the principles of triboelectrification and electrostatic induction to generate electricity or electrical signals. In efforts to augment the electrical output performance of TENGs and broaden their range of applications, researchers have endeavored to refine materials, surface morphology, and structural design. Among them, physical morphological modifications play a pivotal role in enhancing the electrical properties of TENGs by increasing the contact surface area, which can be achieved by building micro-/nano-structures on the surface or inside the friction material. In this review, we summarize the common morphologies of TENGs, categorize the morphologies into surface and internal structures, and elucidate their roles in enhancing the electric output performance of devices. Moreover, we systematically classify the methodologies employed for morphological preparation into physical and chemical

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approaches, thereby furnishing a comprehensive survey of the diverse techniques. Subsequently, typical applications of TENGs with special morphology divided by energy harvesting and self-powered sensors are presented. Finally, an overview of the challenges and future trajectories pertinent to TENGs is conducted. Through this endeavor, the aim of this article is to catalyze the evolution of further strategies for enhancing performance of TENGs.

Keywords: triboelectric nanogenerator, morphology modification, micro-/nano-structures, physical and chemical fabrication, electrical output performance

1. Introduction

Propelled by the relentless evolution of sensor technologies and the burgeoning consciousness surrounding health and wellness, wearable sensors have garnered increasing popularity, significantly enhancing the convenience of daily life [1–3]. However, the reliance of most wearable sensors on conventional battery power poses limitations dictated by the constraints of energy supply. In instances where timely recharging is unattainable, operational efficiency is compromised. To surmount this challenge, many alternative energy solutions are advocated including solar cells, thermoelectric generators, piezoelectric nanogenerators, and triboelectric nanogenerators (TENGs) to engineer self-powered sensor platforms [4–7]. Among these techniques, TENGs have been proved as a promising approach to efficiently convert mechanical energy into electricity or electrical signals [8]. The versatility of TENGs extends beyond their intrinsic power generation capabilities; they can harness ambient mechanical energy, rendering them invaluable for wearable sensor applications [9]. Furthermore, TENG holds promise across diverse domains, spanning from self-powered sensor systems to pioneering energy sources, underpinning its multifaceted utility and relevance.

The first device of TENG was pioneered by Wang's group in 2012 and it exhibits a strong capacity to proficiently convert ambient low-frequency, low-amplitude mechanical energy into electrical power [10]. Triboelectrification is a common phenomenon in daily life. The development of TENGs based on this phenomenon has aroused great interest, as TENGs can produce several hundred volts through triboelectricity, making them a popular energy source. Furthermore, such type of device demonstrates versatility in capturing mechanical energy from diverse sources such as human locomotion, water waves, vibrations, tire rotations, and wind flow, subsequently converting it into electrical energy [11–14]. Typically, TENG comprises electrodes and friction layers structurally, primarily relying on triboelectrification and electrostatic induction mechanisms. In an atom, electrons are distributed around the nucleus in a cloud-like pattern known as electron clouds [15]. When atoms of two materials come into contact, the electron clouds begin to overlap. The overlap causes the electrons to interact more and gain enough energy to cross the energy barrier. This process realizes electron transfer. During contact and sliding, electrons are transferred from the material with lower affinity to the material with higher affinity. Therefore, when two materials with opposite triboelectricity come into physical contact, the material with higher electron affinity acquires a negative charge, while the other surface obtains an equivalent yet opposing positive charge. Subsequent separation of the materials leads to segregation of frictional charges within the interfacial region, inducing a potential difference between the electrodes. Consequently, the movement of free electrons within the electrodes facilitates an alternating flow in an external circuit to maintain electrostatic equilibrium [16].

Currently, TENG has four basic operating modes, including vertical contact-separation mode, lateral sliding mode, single-electrode mode, and freestanding-layer mode [17–20], as shown in figure 1.

Since then, the material, structure, and morphology of TENGs have been explored to enhance the output performance and environmental adaptability for diverse applications [15, 30, 31]. Consequently, various high-performance TENGs suitable for different working environments have been prepared. The electron affinity of a material dictates the polarity and electrical charge generated during triboelectricity interactions. Optimal output performance is typically achieved by pairing two materials with opposing electron affinity characteristics as friction layers. Metal, polymer, textile and biodegradable materials are commonly utilized in the fabrication of high-performance TENGs [10, 32–34].

In addition to selecting materials, a variety of TENGs with different shapes and structures have been designed for environmental adaptation. Based on sliding friction, a gate structure TENG can be utilized for sliding displacement detection in the horizontal direction. A rotating disc structure TENG can detect wind speed by rotating motion and can also be used for human elbow bending detection. Based on rolling friction, TENGs with various structures based on spheres are tailored for fluid monitoring applications [35]. Furthermore, TENGs have been fabricated utilizing fiber textiles to enhance the wearability of the sensor structure [32]. Beyond structural adjustments, morphological modifications of the friction layer can significantly enhance both the output performance and environmental adaptability of the device.

Overall, the output performance of TENGs can be effectively improved through physical and chemical modifications of the material. Chemical modification involves altering a material by introducing specific functional groups to adjust electron affinity or impart unique properties. On the other hand, physical modification, also referred to as topographical modification entails altering the physical structure rather than the material's intrinsic properties [36]. For TENGs, physical modification is accomplished by creating micro-/nano-structures on or within the friction material.

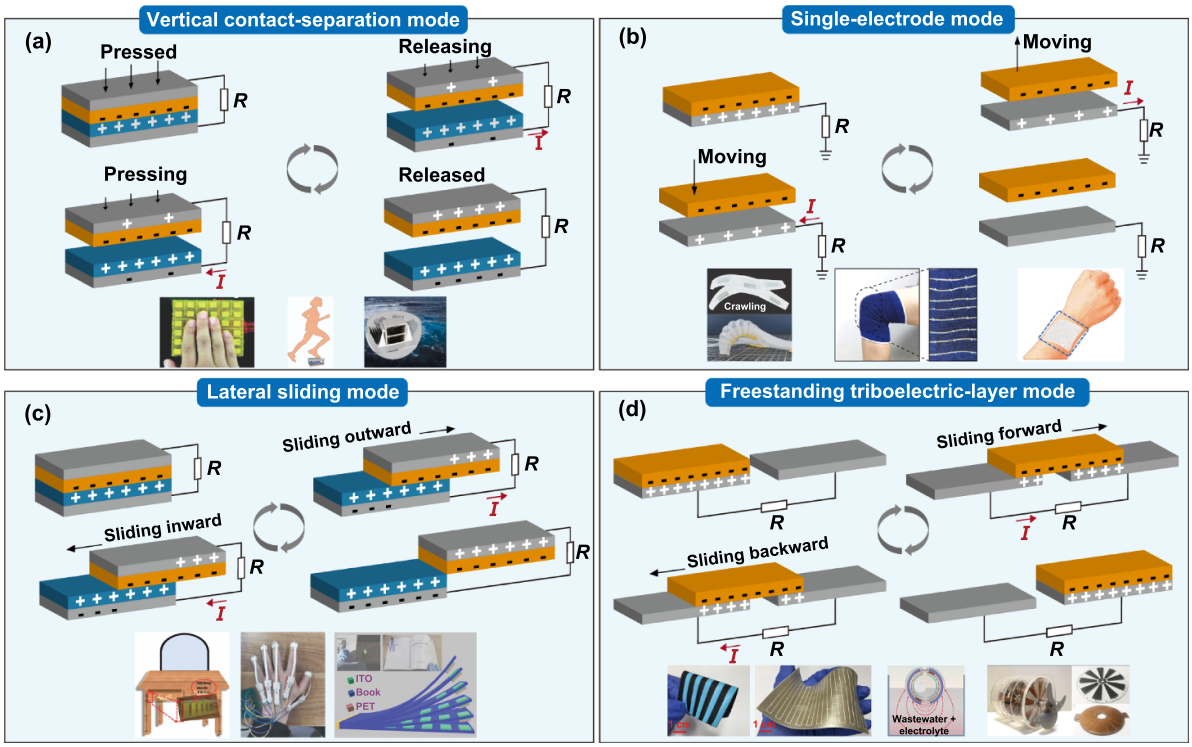


Figure 1. Working process and operating scenario of TENGs in four modes. (a) Vertical contact-separation mode. TENGs with vertical contact-separation mode applied in self-powered shape mapping sensors, sports-monitoring sensors, and blue energy harvesting [21–23]. (b) Single-electrode mode. TENGs with single-electrode mode applied in soft robotic-adapted multimodal sensors, sensors on wearable kneepads, and e-skin [9, 24, 25]. (c) Lateral sliding mode. TENGs with lateral sliding mode applied in table draws for energy harvesting, five-finger motion detection sensors, and page mark paper-based sensors [19, 26, 27]. (d) Freestanding triboelectric-layer mode. TENGs with lateral sliding mode applied in harvesting energy from sports, removal of organic pollutants from water, and ocean energy harvesting [13, 28, 29]. Reproduced from [21] with permission from the Royal Society of Chemistry. Reprinted with permission from [22]. Copyright (2023) American Chemical Society. Reprinted from [23],© 2021 Elsevier Ltd. All rights reserved. [9] John Wiley & Sons. © 2023 Wiley-VCH GmbH. Reprinted from [24],© 2023 Elsevier Ltd. All rights reserved. [25] John Wiley & Sons. © 2023 Wiley-VCH GmbH. Reprinted from [19], © 2022 Elsevier Ltd. All rights reserved. Reproduced from [26]. CC BY 4.0. Reproduced from [27], Copyright © 2014, Tsinghua University Press and Springer-Verlag Berlin Heidelberg. Reprinted from [13], © 2019 Elsevier Ltd. All rights reserved. Reprinted from [28],© 2019 Elsevier Ltd. All rights reserved. Reproduced from [29] with permission from the Royal Society of Chemistry.

These physical morphologies can enhance various surfaces, especially electrical conductivity. For example, micro-/nano-structures of the friction layers can effectively increase the contact area, fundamentally elevating charge density. Materials are selected for the design of TENGs based on the operating environment of devices. Both physical and chemical modification strategies are then integrated to fabricate high-performance TENGs tailored to specific requirements based on the operating environment of devices [37]. In 2012, Wang's group pioneered the application of microscopic geometrical array structures to modify TENGs surfaces, demonstrating the significant enhancement of device output performance through microstructure [38]. The enhancement of TENG output correlates with the enlargement of the effective frictional surface area of the material. Moreover, distinct micro-/nano-structures exert varying impacts on TENG output performance, necessitating to select the suitable morphology and preparation method tailored to specific application requirements.

In this review, we comprehensively summarize the typical morphologies and fabrication methodologies of friction layers for TENG. Firstly, the working principle of TENGs is described, and the role of morphology modification in enhancing performance is explained from the perspective of charge transfer. Then, the morphologies of TENGs are systematically classified and summarized, and their characteristics are compared. They are categorized into microstructure and nanostructure based on the scale of structural units. Based on the shape characteristics of structural units, surface structures are classified into geometric array structures, bionic array structures, wrinkled structures, and other complex patterns.

What's more, the internal porous structure can also be instrumental [39]. It augments both the specific surface area and the degree of compressibility, consequently boosting the output performance. TENG materials exhibit diverse nanostructures such as nano-arrays, nanofiber surfaces, and intricate patterns. Nanostructures can be used for surface modification to create hierarchical architectures atop the microscopic primary structure. Such hierarchical structures have been shown to substantially enhance device sensitivity and expand measurement ranges. The relationship between morphologies of TENGs is shown in figure 2.

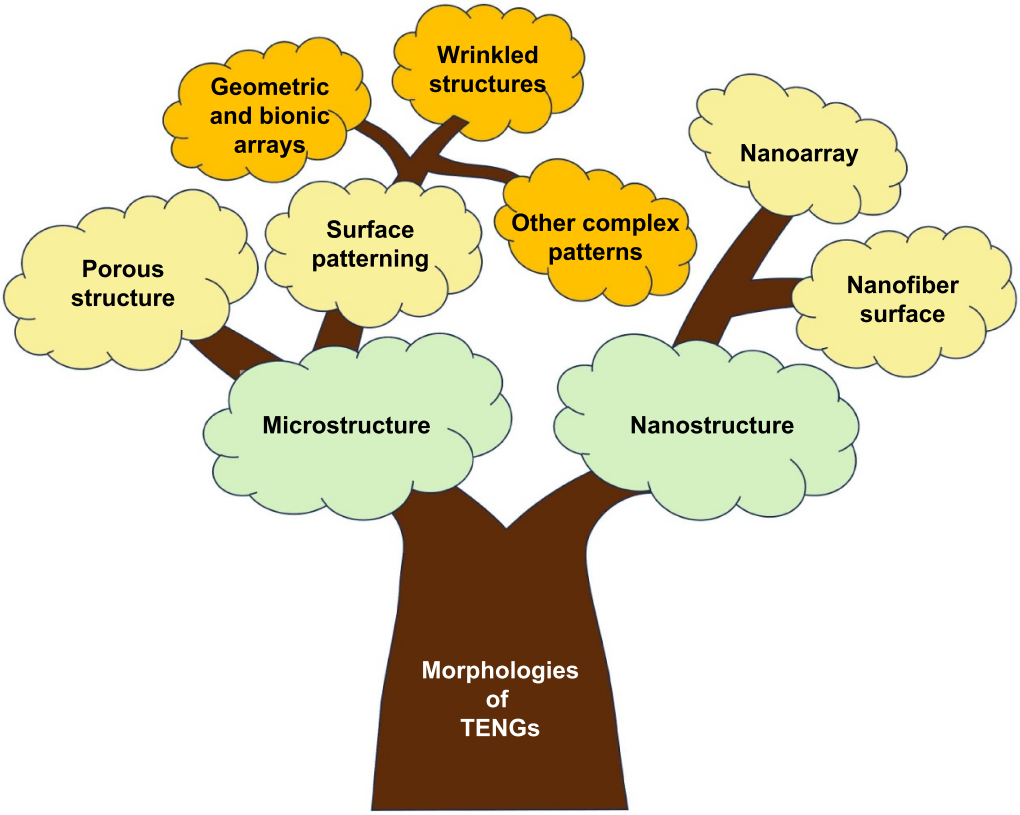


Figure 2. Relationships and classification between morphologies of TENGs.

The morphologies of TENGs are prepared by different methods. Afterwards, we summarize the common preparation methods of TENG morphology. These methods are categorized into physical and chemical methods, based on the principles of morphology preparation. The types of morphology and preparation are summarized in figure 3. In morphology preparation, certain morphologies are prepared by purely physical transformation, encompassing techniques like etching, printing, and sputtering, thus classified as physical morphology preparation processes. Additionally, other methodologies induce chemical transformations, including chemical deposition, chemical etching, and block copolymer (BCP) self-assembly, hence called chemical preparation processes for micro-/nano-structures. Finally, the review encapsulates the application domains of TENG and offers a prospective outlook. The objective of this article is to provide an overview of the morphology and preparation methods of TENGs, foster interdisciplinary collaboration and broaden perspectives of researchers. In summary, the combination of principle, structure, process and outlook is realized.

2. Morphological modification methods and preparation processes for TENG friction materials

It proves that the electrical output performance of TENGs can be improved by morphological modification of friction layers via physical structuring. The phenomenon is mainly attributed to the fact that morphological modification can effectively enhance the specific surface area and roughness of TENGs [60]. Such enhancements can be explained by the charge transfer mechanism underlying TENG operation, which is grounded in frictional initiation and electrostatic induction principles. The theoretical underpinning of TENG charge transfer traces back to Maxwell's equations [61]:

$$\nabla \cdot D = \rho_f \tag{1}$$

$$\nabla \cdot B = 0 \tag{2}$$

$$\nabla \times E = -\frac{\partial B}{\partial t} \tag{3}$$

$$\nabla \times H = J_f + \frac{\partial D}{\partial t}. \tag{4}$$

In the equations, D represents electric displacement field, ρ_f represents the free electric charge density, B represents the magnetic field, E represents electric field, t represents time, H represents magnetizing field, and J_f represents the free electric current density. In equation (4), $\frac{\partial D}{\partial t}$ can be denoted as displacement current J_D, as shown in equation (5):

$$J_D = \frac{\partial D}{\partial t} = \varepsilon \frac{\partial E}{\partial t} + \frac{\partial P_S}{\partial t}. \tag{5}$$

Where ε represents the permittivity of the material, and P_S represents the polarization density of static charges.

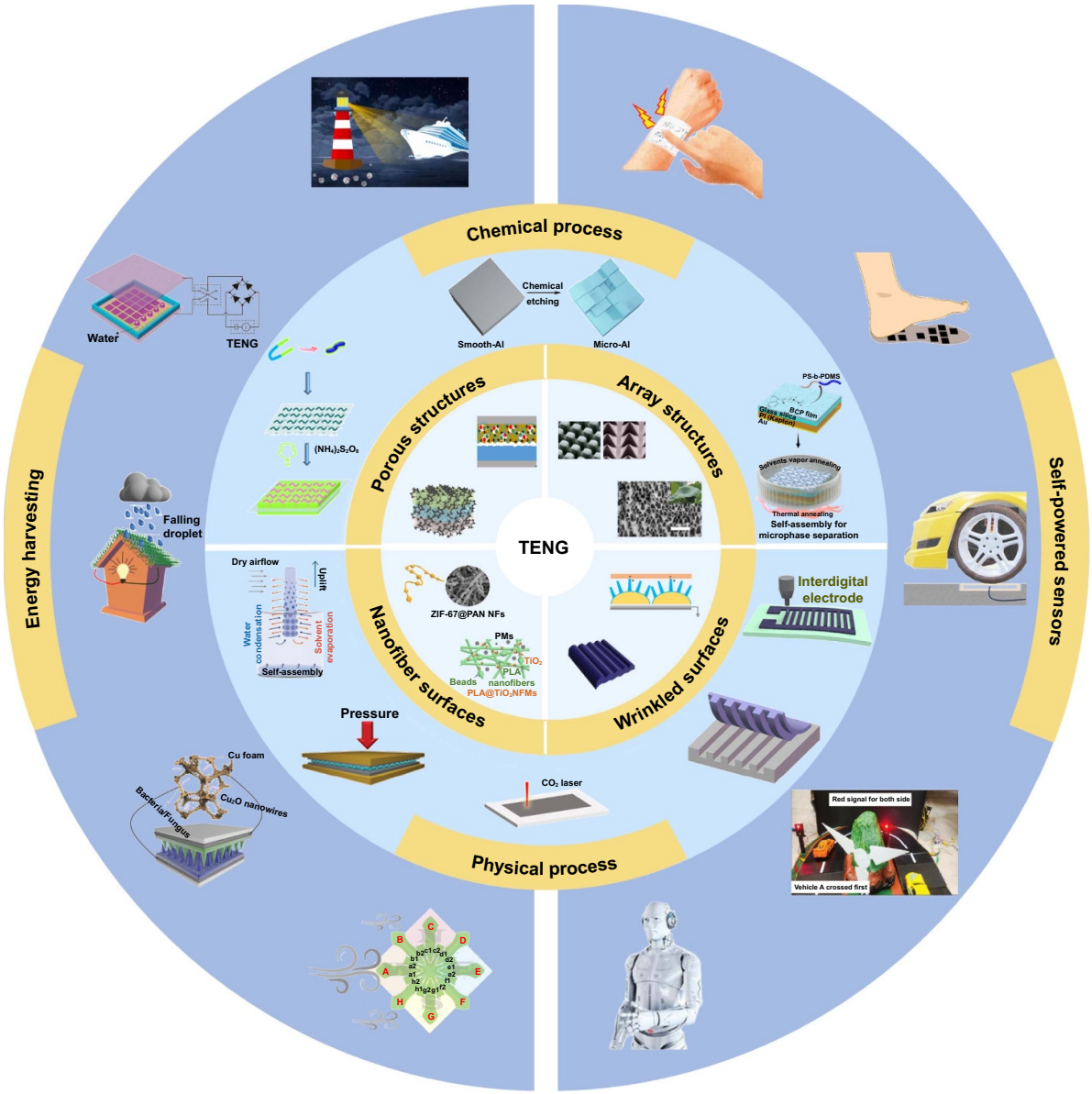


Figure 3. Overview of TENG with special morphologies, including micro-/nano-morphology types, fabrication process, and applications [25, 38, 40–59].

The relationship was derived between the transferred charge (Q), the distance (x) between the two electrodes, and the voltage (V) for TENG operating in the vertical contact-separation mode [62], as shown in equation (6).

$$V = -\frac{Q}{S_{\varepsilon_0}}\left(d_0 + x(t)\right) + \frac{\sigma x(t)}{\varepsilon_0}. \tag{6}$$

In this context, $x(t)$ represents the distance between two frictional layers, σ represents the transferred charge per unit area, ε_0 represents the vacuum permittivity, and d_0 represents the effective dielectric constant.

Equation (6) illustrates that the output performance of TENGs is predominantly contingent upon material properties, and it is directly proportional to both the contact area (S) and the charge density (σ) of friction layers. Consequently, to improve the output performance of the TENGs, it is necessary to increase the effective contact area and charge density of friction layers [63]. The specific surface area of materials accurately describes the effective contact area, which is the surface area per unit volume of solid material. In contrast with flat surfaces, rough surfaces have larger specific surface areas and higher capacitances. Enhancing the surface charge density can increase the potential difference between the two electrodes, thus obtaining an increase in output performance. However, this improvement is costly due to air breakdown limitations. Consequently, researchers usually start from five aspects: the selection of triboelectric materials,

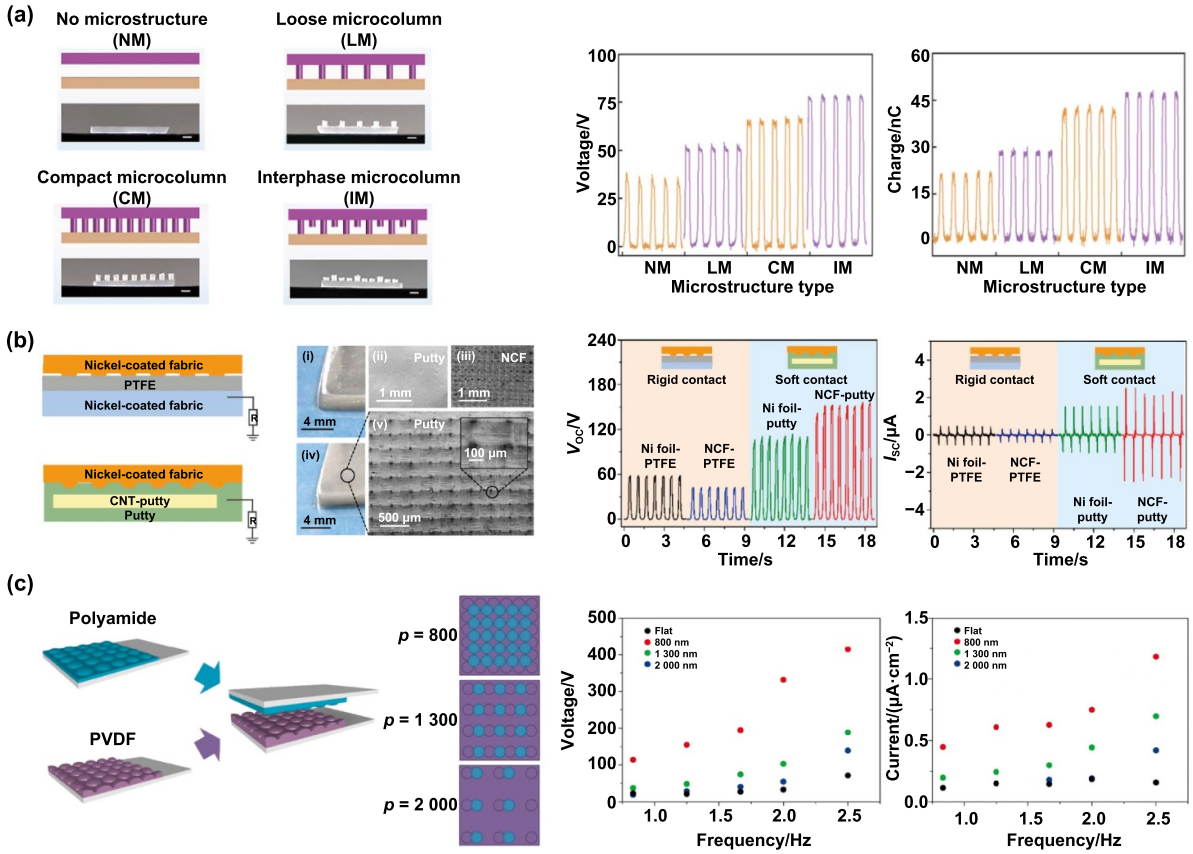


Figure 4. Comparison of TENG performance with and without morphologies, demonstrating the role of morphology in output improvement. (a) The output comparison of TENGs with no microstructure, loose microcolumn, compact microcolumn, and interphase microcolumn. Reprinted with permission from [65]. Copyright (2021) American Chemical Society. (b) The output voltage and current of TENGs based on soft putty and rigid PTEF in contact with planar Ni foil and microrough nickel-coated fabric. Reprinted with permission from [66]. Copyright (2019) American Chemical Society. (c) The voltage and current comparison of TENGs based on polyamide and PVDF with and without semi-ellipsoid array. Reprinted from [67], © 2021 Elsevier Ltd. All rights reserved.

surface modification, material composite, structure design of TENGs, and electrode modification, to enhance the output performance of TENG [64].

Incorporating micro-/nano-structures onto or within TENGs stands as a prevalent strategy aimed at amplifying surface roughness, thereby substantially enhancing the output performance of TENGs. Compared with planar TENGs of the same size, TENGs with micro-/nano-structures can serve higher outputs, as is shown in figures 4(a)–(c). It demonstrates that the output performance of TENGs can be effectively improved by morphological modification. Microstructures include patterned structures on the surface and microporous structures inside. Surface microstructures can assume diverse geometric structures, such as geometric arrays, bionic arrays, and wrinkled surfaces. In addition, a wide range of complex surface patterns can be designed by printing techniques. Microstructures can be obtained by a variety of techniques, such as template method, electrostatic spinning, etching and printing. During the preparation of friction layers, appropriate preparation methods should be selected according to the characteristics and morphology of the material.

2.1. Microstructure

2.1.1. Surface patterning

2.1.1.1. Geometric arrays and bionic arrays. The morphology of microarray structures can be fabricated through a myriad of methods, such as templating, etching, deposition, and electrostatic spinning [68]. Among these methods, the templating method emerges as a common approach for microarray structure fabrication, normally with silicon wafers and polydimethylsiloxane (PDMS) serving as commonly employed materials [69, 70].

Geometric array structure refers to a configuration of regular geometric patterns distributed across the material surface, encompassing various types such as pyramids, inverted pyramids, cylinders, cones, squares, microwires, hemispheres, and nanomounds [64]. All these geometric arrays have demonstrated efficacy in enhancing the output performance of TENGs. Among them, the pyramid arrays are the most widely applied because of obvious improvement in output performance. For preparing geometric arrays, the hard

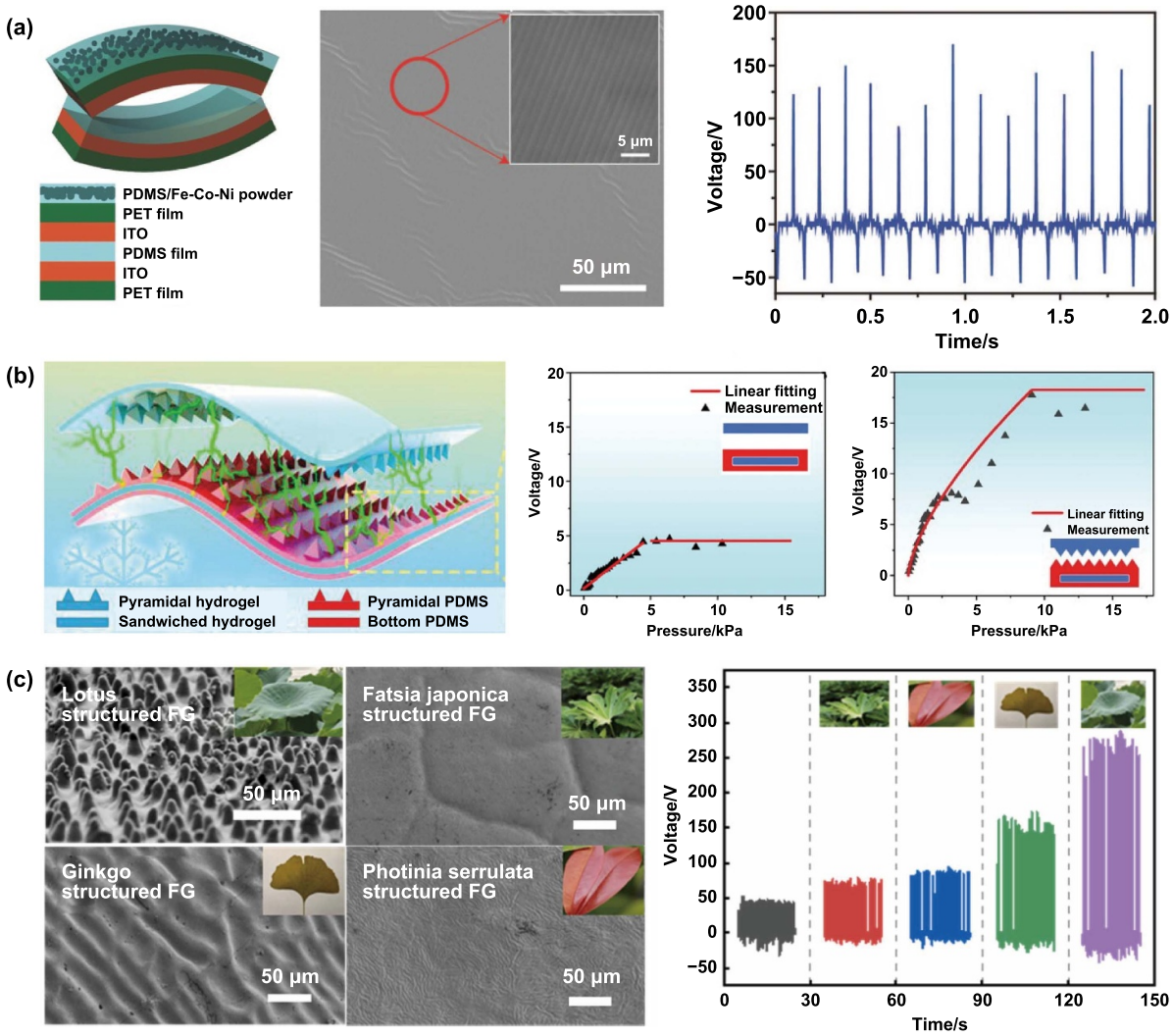


Figure 5. TENGs with geometric and bionic structures. (a) Surface morphology and output performance of TENG with microwire structure. [72] John Wiley & Sons. © 2016 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim. (b) Performance of TENG with micro-pyramid array compared with the plain surface. Reproduced from [73]. CC BY 4.0, demonstrating the enhancement of output performance by morphology. (c) The bionic morphology and output voltages of four-leaf structures. Reprinted from [40], © 2023 Elsevier Ltd. All rights reserved.

template method is a common approach. In 2012, Wang's group pioneered the application of regular geometric arrays on friction layers to enhance the electrical output performance of TENGs [38]. The hard template method was utilized to create microscopic array patterns on silicon wafer molds by etching. Subsequently, relative patterns were transferred onto PDMS using the molding-demolding method, yielding three distinct geometrical arrays: line, cube, and pyramid arrays. The output of TENGs with surface geometric arrays consistently outperformed those based on flat PDMS films, demonstrating the positive effect of geometric array modification on enhancing TENG output efficiency. Compared to the TENGs based on flat PDMS counterparts of equivalent dimensions, the output of TENGs modified with pyramid or cube structures exhibited approximately 5–6 times higher. The TENG with pyramid structures of 4.5 cm $\times$ 1.2 cm could generate a maximum output voltage and output current of 18 V and 0.7 μA. Furthermore, sandpaper and discs could serve as a template for replicating surface morphology [71, 72], as shown in figure 5(a). Morphology transfer techniques were used to fabricate resembling geometric structures on the surface of TENGs.

Based on the previous studies, many methods have been improved to achieve geometric array morphology preparation of a variety of material surfaces. A TENG based on double-network (DN) hydrogel enabled tactile sensing. Micro-pyramid structures on the surface could effectively enhance the sensitivity of tactility, as shown in figure 5(b). Micro-pyramid structures were constructed on the surface of an ionic DN hydrogel, and a pyramid-patterned hydrogel fabrication method and laminated PDMS encapsulation process were used. In the two-electrode TENG modes, lithium bromide immersion treatment hydrogel and PDMS were used as the friction layers, and micro-pyramid arrays were distributed on the surface of both layers. Similarly, patterned silicon wafers were used as templates to prepare micro-pyramid

structures. N′-Methylenebis(acrylamide), ammonium persulfate, KCl, and carrageenan were utilized to prepare DN ionic organic hydrogels. Multilayered hydrogel structures with micro-pyramid morphology were prepared by spin-coating, molding, peeling and final assembly processes. In addition, PDMS layers with pyramid arrays were prepared in the same method. In the case of the two-electrode TENG, the output saturation voltage was 4.5 V for plain PDMS and plain hydrogel when subjected to a saturation pressure of 4.5 kPa. However, when PDMS with micro-pyramids and hydrogel with pyramid morphology were employed, the output saturation voltage increased to 8.5 V. The micro-pyramid morphology contributed to an enlarged contact area and heightened triboelectric effect, thus enhancing output performance [73, 74].

The electrostatic spinning process also provides a flexible approach for preparing structured arrays. Far-field electrostatic spinning could be used to fabricate micro-pyramid arrays on fabric surfaces [75]. An electrically grounded aluminium foil served as the collector. At the same time, poly(vinylidene fluoride) (PVDF) functioned as the spinning material, enabling precise control over voltage, humidity, and horizontal swinging distance to regulate pyramid structure growth. The PVDF jets became positively charged during this process, attracting negatively charged microdomains with higher PVDF deposition thickness on the collector surface. This preferential deposition led to the formation of pyramid structures. Moreover, adjacent pyramids could merge during spinning, facilitating the gradual growth of pyramid structures from the nanometer to micron or even millimeter scale.

Apart from silicon wafers as hard templates, plant and animal tissues as well as PDMS have been explored to fabricate geometric array structures. This exploration aims to produce bionic array structures resembling geometric arrays [76]. These surface structures emulate the morphology of natural tissues, offering environmental benefits and a straightforward fabrication process. Templates were selected in past studies such as insect wings, flower petals, and plant leaves to achieve geometrically analogous surfaces [16, 77, 78].

In addition, gelatin sourced from fish scales is biodegradable. For the eco-friendliness of devices, gelatin with leaf surface microstructures could be developed into a TENG [40]. The surface structures are shown in figure 5(c). This approach offered simplicity in fabrication, utilizing materials from natural sources, thereby promoting cost-effectiveness and facile degradation. Surface morphologies of lotus, fatsia japonica, photinia serrulata, and ginkgo leaves resembled pyramid, square, groove, and grating array structures, respectively. The microstructures were transferred from leaf surfaces to fish gelatin via sequential negative and positive mold preparations. Subsequently, PDMS precursor was cast onto the molds and cured into films to replicate the surface structures. Flexible PDMS molds were then cleaned and filled with fish gelatin, dried, and demolded to yield friction layers mirroring leaf surfaces. Among these, the TENG featuring a pyramid-like array surface emulating lotus leaf exhibited the highest performance, generating a maximum output voltage of 320 V and current of 0.80 μA.

2.1.1.2. Wrinkled structure. Apart from geometric and bionic array structures, wrinkled structures are also a prevalent type of surface microstructure. Unlike other morphologies, wrinkled structures offer good ductility and tensile properties, ensuring stable electrical performance within a certain strain range. These structures can be achieved through stretching, etching, extrusion-based printing, fluorocarbon plasma treatment, and heat treatment methods [64]. The simplicity and cost-effectiveness of these processes facilitate the large-scale manufacturing of wrinkled structures. A multifunctional liquid metal electronic skin with wrinkled surfaces was developed, and the wrinkled structures were fabricated by stretching and releasing [25]. The electronic skin was based on the triboelectricity principle and could harvest mechanical energy, as shown in figure 6(a). The material was pre-stretched and activated by repeated uniaxial stretching, resulting in transverse mesh and longitudinal wrinkled structures on a fiber mat coated with liquid metal.

Furthermore, the wrinkled structure is often combined with other morphologies, enabling the device to enhance performance. Hierarchical wrinkled surfaces are composed of micro-structures and nano-structures. Therefore, TENGs with hierarchical wrinkled surfaces can detect signals in different intensities and ranges. A highly sensitive and broadly detectable e-skin was based on triboelectricity, which was composed of layered wrinkled membranes of PDMS/ZnO nanowires and fluorinated ethylene propylene (FEP). The special morphology enabled the liquid-metal e-skin to be prepared for sensors with high sensitivity and a wide range [41]. The preparation of wrinkled films utilized the principle of linear expansion, thus the film appeared wrinkled surface due to uneven thermal expansion. A layer of ZnO was deposited on the surface of the PDMS film and heat-treated. Due to the mismatch of thermal expansion coefficients between PDMS and ZnO, the top surface of the PDMS/ZnO hybrid structure tended to shrink locally, spontaneously forming wrinkled structures. Graded wrinkled films were obtained by depositing a layer of ZnO nanowires (ZnO NWs) on the surface of the wrinkled films. The piezoelectric potential from the deformation of ZnO nanowires significantly enhanced the frictional charge transfer capability of TENGs, while the graded structure facilitated the establishment of a dynamically self-regulated contact region. The surface and output performance are shown in figure 6(b).

2.1.1.3. Other complex patterns. In addition to array and wrinkled structures, other complex patterns can be fabricated by various printing methods. The patterns constructed by printing are 2D and 3D configurations. Inkjet printing, spraying, screen printing, electrostatic spinning, and extrusion-based printing are employed for constructing 2D patterns on TENG surfaces [60, 79–83]. Moreover, techniques like spraying, electrostatic spinning, and extrusion-based printing (direct ink writing) enabled the preparation of 3D patterns on TENG surfaces [79]. Three kinds of complex pattern surfaces are shown in figure 7.

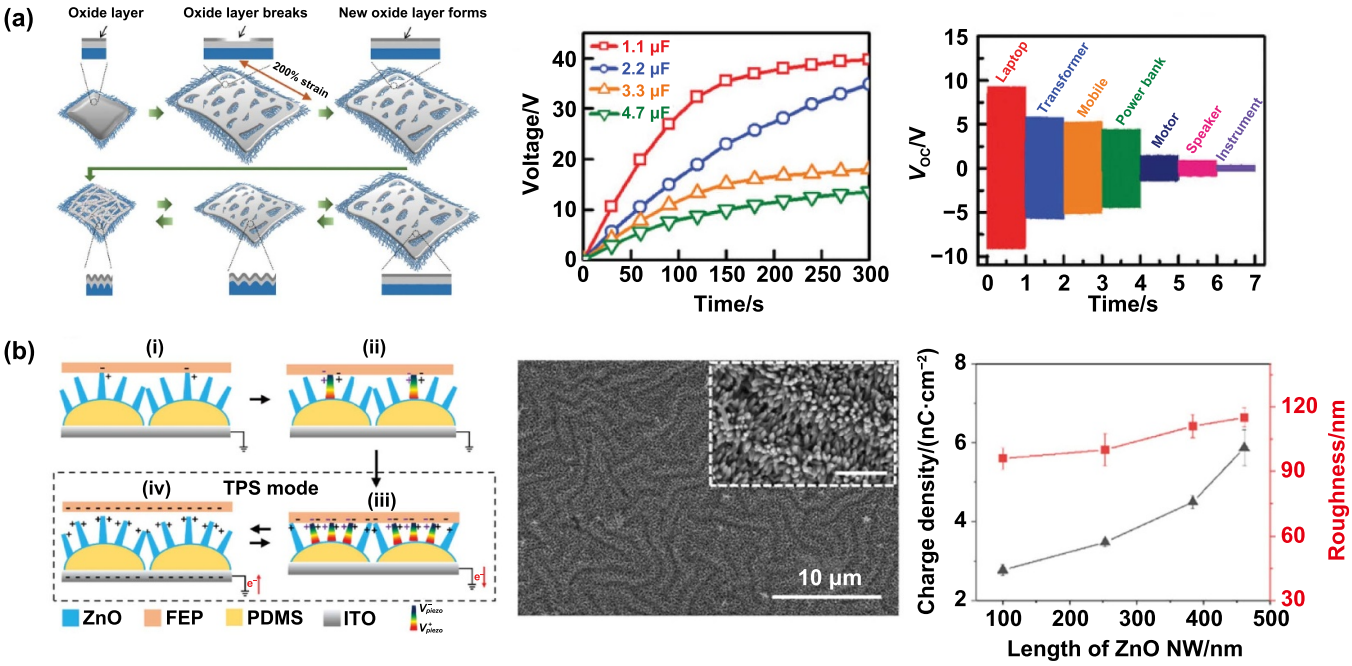


Figure 6. Surface morphology and output performance of TENG based on wrinkled surfaces. (a) The process of wrinkled surface preparation by uniaxial stretching and output performance of the device. [25] John Wiley & Sons. © 2023 Wiley-VCH GmbH. (b) PDMS/ZnO NWs graded wrinkled surface and the trend of charge density and roughness with the length of ZnO NWs. Reprinted with permission from [41]. Copyright (2024) American Chemical Society.

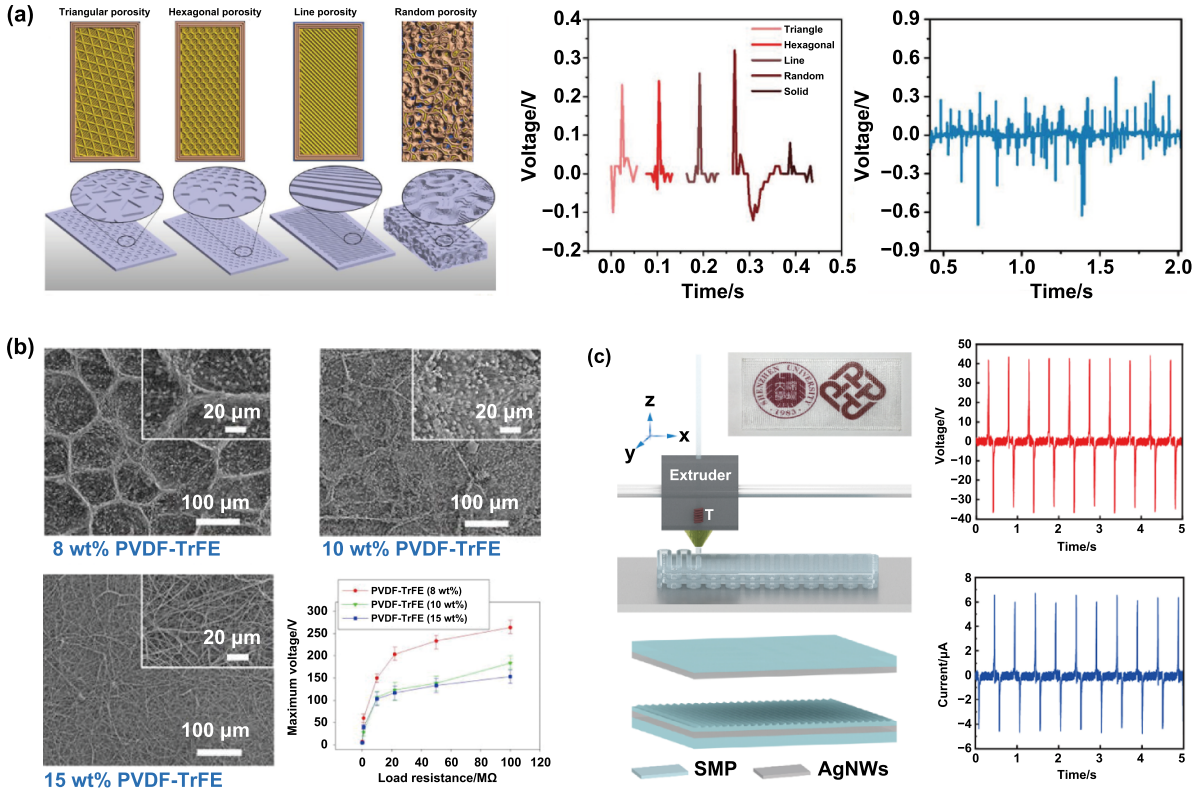


Figure 7. Special morphology of complex patterns. (a) Four kinds of complex patterns generated by 3D printing techniques and their output properties. [54] John Wiley & Sons. © 2022 Wiley-VCH GmbH. (b) Honeycomb-like structures generated by spinning techniques and their output properties. Reprinted from [84], with the permission of AIP Publishing. (c) Microwires prepared by extrusion-based printing methods. Reprinted from [85], © 2021 Elsevier Ltd. All rights reserved.

A porous 3D-structured TENG was fabricated by extrusion-based printing [54]. Initially, planar substrates featuring linear, triangular, hexagonal, and random porous structures were 3D printed from polylactic acid (PLA) filament. Subsequently, a thin layer of Gd_2Te_3 was coated onto the substrates. The porous structure significantly augmented the surface area, leading to a remarkable increase in TENG output efficiency by approximately 400%, as is shown in figure 7(a).

The 2D honeycomb structure was composed of nanofibers, and was fabricated by electrostatic spinning self-assembly with polyacrylonitrile, polyvinyl alcohol (PVA), and poly(ethylene oxide) [84]. The resulting fibrous TENG exhibited a promising power density of 1.6 $W{\cdot}m^{-2}$. The performance of TENGs with honeycomb surface structures surpassed those with flat nanofiber surfaces by more than two-fold, as shown in figure 7(b).

Moreover, Huang's group devised microwire surfaces employing shape-memory polymer (SMP) filaments through fused deposition by 4D printing, an extrusion-based method leveraging shape-memory materials [85], as is shown in figure 7(c).

2.1.2. Internal and surface porous structure. The pore sizes in porous materials are typically categorized into three main classes: micropores (pore size less than 2 nm), mesopores (pore size between 2 nm and 50 nm), and macropores (pore size greater than 50 nm) [39]. Porous materials are widely used in the field of TENGs to generate friction charges on the surface of pores deep in the friction layers. The introduction of porous structure in friction electric materials can effectively increase the contact area and improve the charge density. Porous materials are widely used in portable sensors due to their lightweight and admirable mechanical properties [22].

Porosity significantly impacts the mechanical and electrical characteristics of materials. Higher porosity enables greater storage of conductive material, thereby augmenting the sensitivity and energy conversion efficiency of the device [86]. When pressure is applied or released, the framework of porous structures can act as air gaps during contact or separation. A porous TENG can be seen as an integrated assembly of multiple micro TENGs, offering a larger contact area compared to other structures. Various TENGs based on porous materials have been developed, including aerogels, hydrogels, foams, and fibers. Common methods for their preparation include the template method, freeze-drying, carbonization, foaming, template assembly, chemical deposition, electrostatic spinning, and 3D printing. Porous structures distributed both on the surface and inside the friction layers can enhance the output performance. A tube-shaped TENG was fabricated by planar aluminum foil with porous surface nanostructures and a hemispherical PVA hydrogel [87]. Motions such as flexion, torsion, and extension of the elbow joint could be detected by this TENG. A TENG based on aerogel films had a porous structure and was prepared by freeze-drying method [88]. These TENGs utilized porous polymers as positive and negative friction layers, employing porous chitosan aerogel films (86% porosity) and porous polyimide (92% porosity) as the friction layers. The internal porous structure allowed the materials to induce charges not only on the contacting surfaces but also on the porous surfaces upon contact. The porous polymer film exhibited greater deformation and thus higher capacitance compared to dense films. Aerogel-based TENG demonstrated a wide measurement range, high sensitivity, and excellent mechanical properties, generating a voltage of 60.0 V and a current of 7.7 μA with a power density of 2.33 $W{\cdot}m^{-2}$. The three steps of grafting, complexation and salting out could improve the electrical and mechanical properties of PVA hydrogels [57]. The hydrogels exhibited remarkable damage resistance, boasting a tensile stress of 6.47 MPa, a strain of 1000%, a toughness of 35.24 $MJ{\cdot}m^{-3}$, and a fracture energy of 37.59 $kJ{\cdot}m^{-2}$. The robust crystalline PVA domains and reversible/noncovalent interactions contributed to its high-density crosslinked network structure. The nanoscale internal pore arrangement of the crystals was suitable for TENG applications. This hydrogel-based TENG achieved an open-circuit voltage of 150 V and a short-circuit current of 3.0 μA. The microporous structure could also be obtained by the breath figure method. Silver-plated nylon fibers spun into honeycomb porous microstructures could be utilized for textile TENGs. The pore size induced by the breath figure method varied with the axial gradient [50]. This process entailed three main stages: nucleation, growth, and self-assembly of the HPM arrays. Initially, the fiber substrate was dip-coated with SBS/$CHCl_3$ solution, leveraging the elastomeric properties of SBS, a triblock copolymer, to enhance mechanical resilience. Subsequently, the coated fibers were suspended in a high-humidity environment, causing moisture from the volatile $CHCl_3$ to condense on the fiber surface, initiating nucleation. As the fibers were gradually raised from the container, evaporation of $CHCl_3$ induced simultaneous drying of the outer fiber surface and continued condensation of moisture within the container, leading to a gradient change in pore size along the fiber length, culminating in the formation of the desired HPM upon complete extraction. TENG based on multifunctional silver-plated nylon fibers surface-engineered with HPMs with an area of 2.5 cm $\times$ 2.5 cm were able to produce a power density of 390.8 $mW{\cdot}m^{-2}$.

Sensor applications in diverse scenarios necessitate varying levels of sensitivity and tuning ranges, which are influenced by factors such as pore size, porosity, and pore distribution within the material. The method to regulate pore sizes based on self-assembly was proposed for a porous patch TENG, as shown in figure 8(a). This TENG was crafted using a 30% silicone rubber dilution for the porous negative friction layer, while graphene served as the electrode and positive friction layer. The porous friction layer employed PU sponges as scaffolds, immersed in a diluted solution of silicone rubber (Ecoflex-00-30) containing ethyl acetate. Following complete absorption of the diluent, curing of the sponge was carried out in an oven. Subsequent heat-induced volatilization of ethyl acetate left behind only cross-linked silicone rubber within the pores, forming a porous dielectric layer. The volume of silicone rubber retained in the pores depended on the silicone

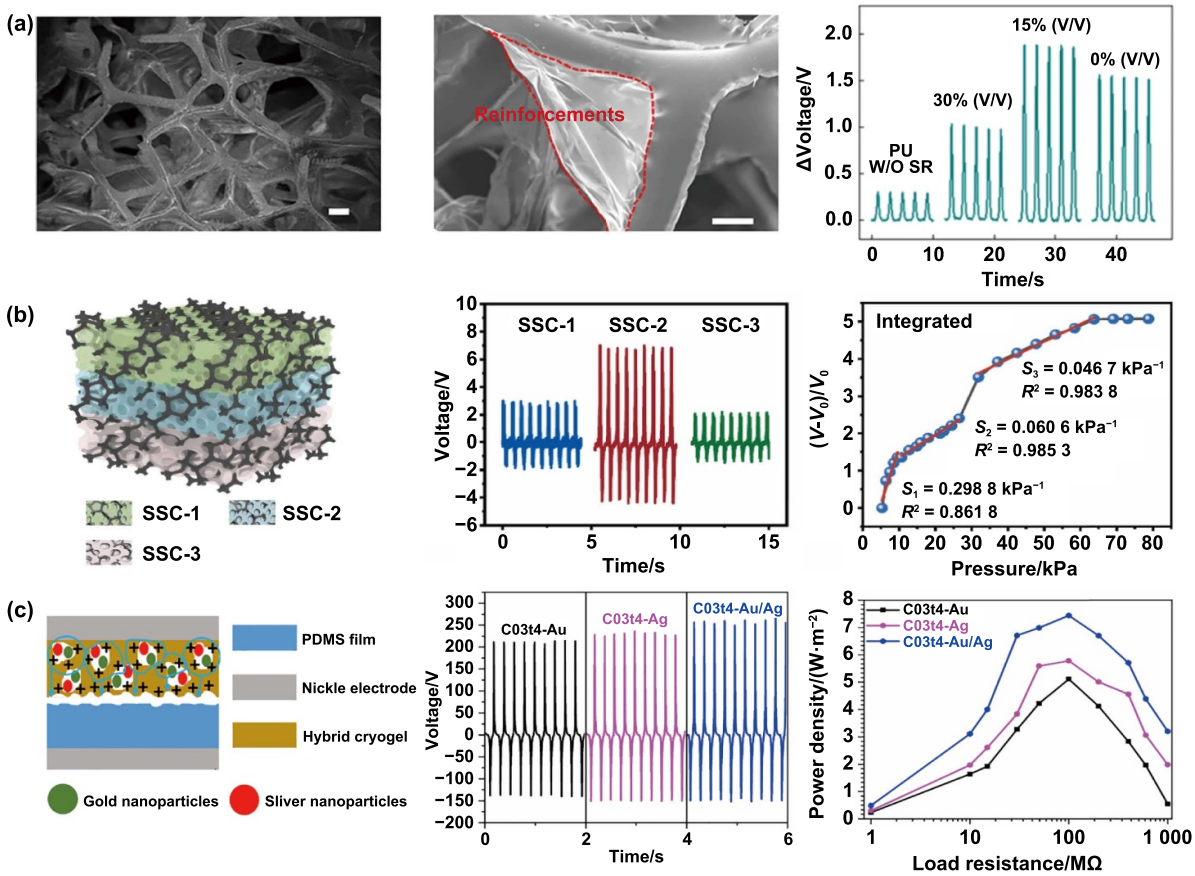


Figure 8. Surface and internal porous structures and porous TENG output performances. (a) TENG made of porous PU sponges with Ecoflex attachment. Reproduced from [90]. CC BY 4.0. (b) Structure and output performance of single-electrode TENG with hierarchical porous structure. Reprinted from [45], © 2024 Elsevier Ltd. All rights reserved. (c) Porous TENG with silver and gold nanoparticles. [46] John Wiley & Sons. © 2020 John Wiley & Sons Ltd.

rubber content in the diluent, offering a means to finely adjust pore size by regulating the concentration of the silicone rubber solution, thereby modulating the material's mechanical properties. Notably, a sensor unit sized at 1.5 cm × 1.5 cm exhibited a detection range of up to 50 kPa and a sensitivity of 0.21 kPa^{-1} [89, 90]. A single-electrode TENG using conductive polyurethane (PU) sponges and foamed silicone rubber was tailored for smart insoles aiming at motion monitoring and foot disease diagnosis, as is shown in figure 8(b). They immersed carbon nanotube-coated PU sponges into various ratios of chemically foamed silicone rubber precursor solutions, resulting in three monolayer composites with distinct porosities. These composites were then integrated to form a graded porous sensor endowed with gradient compression properties, showcasing a sensitivity of 0.2988 kPa^{-1} within the low-pressure range and a maximum detection pressure of 63.75 kPa [45].

Furthermore, based on the porous structures, Wang's group utilized a composite of nanoparticles and a microporous framework to augment output performance, introducing a microporous hole-nanoparticle composite structure [21]. PDMS, gold nanoparticles, and water were blended. Due to the hydrophobic nature of PDMS, PDMS developed pore structures with varied sizes and positions dispersed across the material surface after drying. Upon the application of external force, charge transfer occurred between PDMS and gold nanoparticles within each cavity, thereby inducing electric current generation.

A highly porous hybrid bimetallic cryogel was fabricated to enhance the performance of TENGs [46], as shown in figure 8(c). The TENG device was constructed using a porous hybridized cryogel film composed of silver and gold nanoparticles as friction-positive materials, and PDMS as friction-negative materials. With dimensions of 1 cm × 2 cm, the device achieved an output voltage of 262.14 V and a current density of 27.52 mA·m^{-2}, along with a peak power density of 7.44 W·m^{-2}.

2.2. Nanostructure

2.2.1. Nanoarray. Fabricating nanoarray on the friction layers proves instrumental in enhancing material roughness. Nanoarray, albeit slightly less ordered than microarray, offers a finer scale. Nanoarrays can be crafted through surface reshaping by the template method or by nano-materials attachment, such as nanoparticles, nanotubes and nanoplates [91–93]. The generation process of nanotube patterns on the polyethylene (PE) film by alumina templates is shown in

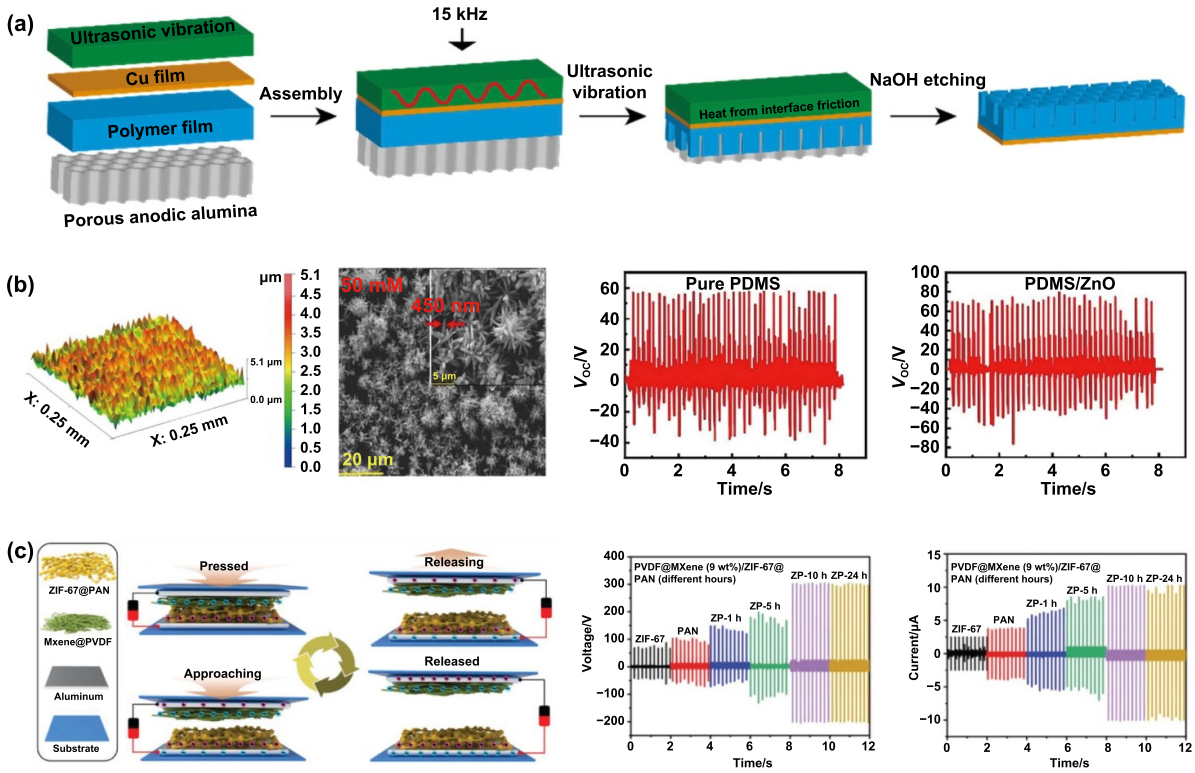


Figure 9. TENG surfaces with nano-structure. (a) Preparation process of polymer nanowires. Reprinted with permission from [94]. Copyright (2020) American Chemical Society. (b) Morphology of PDMS surface with ZnO nanowires and voltage comparisons demonstrating the enhancement of nanostructures for output performance. Reprinted from [96], © 2023 Elsevier Ltd. All rights reserved. (c) Structure and output of TENG with pearl-necklace-like fibers. [44] John Wiley & Sons. © 2023 Wiley-VCH GmbH.

figure 9(a). The PE film was assembled with porous alumina and subjected to 120 °C heating for 30 min. The resultant melting of the PE film enabled its integration into the porous alumina structure, yielding nanotube arrays. Subsequent release of the alumina templates was achieved through NaOH solution etching [94]. Afterwards, nanowire structures were prepared on PVDF surfaces using porous anodic aluminum oxide as a template by the ultrasonic-assisted process [95]. Ultrasonic waves could induce friction between multilayer materials to generate high temperatures, causing PVDF to melt and enter the nano-channels of porous aluminum. The nanowire surface structures were obtained after demolding. The ultrasonic-assisted template method was a controllable and universally applicable method for nanowire preparation. The PVDF-based TENGs prepared based on this technique were capable of generating a voltage of 153 V and a current output of 23 μA.

Moreover, ZnO nanotubes could be attached to the PDMS by chemical bath deposition [96]. Initially, the PDMS-coated substrate was immersed in zinc acetate aqueous solution to facilitate the formation of a ZnO seed layer. Subsequently, the substrate was transferred into a Zn^{2+} aqueous solution to initiate the growth of ZnO nanotubes. The resulting single-electrode TENG, fabricated using PDMS modified with ZnO nanotubes, exhibited an output voltage of 87.5 V, a current of 17.9 μA, and a maximum power density of 36.16 μW·cm^{-2}. Figure 9(b) shows the morphology of the PDMS surface with ZnO nanowires and the output performance.

Apart from dense structures, porous TENG can also enhance the output performance through the addition of nano-arrays. One-dimensional carbon nanotubes (CNTs) and two-dimensional boron nitride particles were mixed in a porous foam [97]. This composite was formed by mixing boron nitride particles and CNTs with N'N'-dimethylformamide and inducing porosity via the supercritical carbon dioxide method. When deformation occurred, the contact-separation process of exposed boron nitride particles and CNTs generated electrical signals. The average output of all-in-one structured TENG foam with an area of 1 cm^2 was around 70 nA and 17 V.

2.2.2. Nanofiber surfaces and patterns. Nanofibers, similar to nanowires, represent one-dimensional nanostructures characterized by their nano-sized radial dimension and unrestricted lateral size. Techniques like electrostatic spinning enable the controlled fabrication of nanofibers with varying lengths, thus yielding diverse surface patterns. Materials suitable for nanofiber production typically exhibit a requisite level of abrasion resistance, such as polyimide and PVDF [98].

The construction of various surface patterns using nanofibers is tailored to meet the specific performance requirements of the device. Critical factors influencing fiber morphology include the concentration, viscosity, and surface tension of the precursor solution. Lower concentration solutions

yield reduced viscosity, leading to jet breakage during electrospray or electrostatic spinning, resulting in fiber formation with beads. With increasing concentration, viscosity rises, causing bead shapes to elongate from spherical to fusiform, and promoting the formation of uniform beadless nanofibers due to enhanced chain entanglement and viscosity. Further concentration escalation results in progressively thicker uniform nanofibers. For homogeneous nanofibers, they can attach to the surface of the material and prepare a surface pattern that enhances the surface roughness and contact area [68, 99]. Nanofibers were spun into the shape of honeycomb for TENGs. The honeycomb network was obtained by assembling poly(vinylidene fluoride-co-trifluoroethylene) on the aluminum foil [84].

Furthermore, by precise control of material selection and electrospinning parameters, specialized structures such as spun microspheres, hollow fibers, and 3D aerogels could be fabricated. In the electrospinning-electrospray process, the PLA nanofibers with TiO_2 nanoparticles could form bead-on-string structures by controlling humidity [43]. TiO_2 dispersions were generated by dispersing nanoparticles in water using plasma technology. Different spinning solutions were formulated by separately adding PLA spinning solution and TiO_2 dispersion into the syringe while adjusting the flow rate. The spinning solution had low viscosity and formed a jet during the spinning process. In high humidity, reduced evaporation rates hindered filament stretching, resulting in prolonged filament connections between droplets, thus forming bead-on-string structures.

Polyacrylonitrile nanofibers were electrospun for TENGs and modified with zeolitic imidazole framework-67 nanocrystals [44]. The process is shown in figure 9(c). Polyacrylonitrile nanofibers were electrospun to form a fiber film and modified with zeolitic imidazole framework-67 nanocrystals. The film was used as the friction layer of TENGs. The prepared nanofibers were soaked in ZIF-67 solution to make the nanocrystals attached. Pearl-necklace-like fibers capable of being applied to TENG were prepared by electrospinning and self-assembly. The wetland weed Juncus effusus *L*. could extract hollow stellate cellulose for TENG applications [100]. The fabrication involved delignification, dialysis, and natural drying processes to yield hollow stellate cellulose units. These units exhibited enhanced surface roughness and porosity, thereby augmenting the electric properties. Another hollow fiber was fabricated, which was filled with liquid metal. This multifunctional fiber had outstanding stretchability and could generate triboelectric signals. In addition, the induced electrification of the liquid metal enabled the generation of electrical signals [101]. The fiber was fabricated using poly(styrene-b-(ethylene-co-butylene)-b-styrene) (SEBS), known for its softness and suitability for melt extrusion. SEBS resin pellets were melted at 230 °C, extruded through a hollow fiber die, and subsequently stretched. The resulting hollow fibers were then cooled in a water bath. Multifunctional fibers were fabricated by injecting eutectic gallium–indium metal liquid into the fibers using a syringe. These fibers exhibited commendable tensile properties. Alginate aerogel and polysuccinimide (PSI) nanofibers were incorporated for TENGs [102]. The PSI nanofiber films were fabricated through electrostatic spinning, followed by ultrasonication for 5 min to produce short nanofibers of micron length. These nanofibers were then dispersed into the alginate solution, and $CaCl_2$ solution was added to form hydrogels. By freezing the hydrogel with liquid nitrogen, the aerogel containing nanofibers was obtained. Utilization of this aerogel in TENGs was found to enhance biocompatibility and mechanical durability significantly. The output comparison of TENGs with different morphologies are summarized in table 1.

2.3. TENG friction material morphology preparation process

Herein, we categorize physical topographical processes into chemical and physical preparation methods depending on the process types. It is important to highlight that, unlike chemical modifications, chemical preparation methods for surface topography solely encompass chemical processes in constructing the new topography (such as attaching nanoparticles to the material surface), without altering the molecular structure of the material. The characteristics of different processes for morphology preparation are summarized in table 2.

2.3.1. Chemical preparation processes. Chemical preparation methods utilize chemical reagents and processes to create micro-/nano-structure morphologies, often involving liquid chemical reagents. These methods are frequently employed to fabricate nanostructures on surfaces. Typical chemical modification techniques include chemical deposition, chemical etching, and BCP self-assembly.

Chemical deposition method is a common additive modification, which has the advantages of thickness controllability, high reproducibility and low cost. Chemical deposition can be utilized on metals, semiconductors and polymers, and is classified into liquid-phase deposition and gas-phase deposition depending on the state of the precursor. In gas-phase deposition, a gaseous precursor substance is introduced into a reaction chamber, leading to the deposition of the desired material onto the substrate surface through a chemical reaction. Liquid-phase deposition is more frequently employed for surface morphology preparation, involving the use of chemical solutions (like metal salts, metal-organic compounds, or other reagents) as precursors, with the substrate typically immersed in the solution.

The desired structures are deposited on the substrate surface via controlled chemical reactions, modulated by factors such as temperature, pH, solution concentration, and immersion duration. Chemical deposition offers a straightforward and cost-effective approach commonly employed for crafting surface nanostructures like nanoparticles and nanowires. The process of fabricating a single-electrode TENG with functionalizing ZnO nanotubes is shown in figure 10(a) [96]. This involved growing ZnO nanotubes onto a PDMS substrate via a simple chemical bath deposition method. Initially, the PDMS precursor was spin-coated onto an ITO glass surface. The PDMS film was obtained after drying. Subsequently, a two-step chemical deposition process was employed: first, a ZnO

Table 1. Output comparison of TENGs with different morphologies.

Morphology	Characteristic	Preparation methods	Triboelectric material 1	Triboelectric material 2	Output performance	References
Geometric arrays	Surface structures, geometric units neatly arranged, widely applied	Template method, etching, deposition, electrospinning, self-assembly, imprinting	PET	PDMS	18 V, 0.7 μA	[38]
			Hydrogel	PDMS	18 V, 20 μW·cm^{-2}	[73]
			Au	PVDF	27 V, 5.5 mA·m^{-2}	[75]
Bionic arrays	Surface structures, similar to geometric structures, molded on plant and animal tissues	Template method	Skin	PDMS	56 V, 3.1 μA	[40]
			Fish gelatin	PDMS	320 V, 0.8 μA	[78]
Wrinkled structures	Pleated structures on the surface	Stretching, etching, and heat treatment methods, printing	Polyurethane	Silicone rubber	205.6 V, 2.3 μA	[25]
			ZnO	FEP	0.26 nC·cm^{-2}·kPa^{-1} (1 kPa—25 kPa)	[41]
Other complex patterns	2D and 3D patterns on surfaces	Electrospinning, printing	Al	Poly(vinylidene fluoride-co-trifluoroethylene) nanofiber	160 V, 17 μA	[84]
			Gd_2Te_3	Water drops	0.5 V, 2.2 μA·cm^{-2}	[54]
Porous structures	Surface and internal structure	Template method, freeze-drying, carbonization, foaming, deposition, electrospinning, 3D printing, breath figure method	Chitosan aerogel	Polyimide (PI)	60.6 V, 7.7 μA	[88]
			Latex	Ecoflex, Hydrogel	150 V, 3.0 μA	[57]
			Polystyrene-block-Polybutadiene-block-Polystyrene	PTFE	97.3 V, 3.0 μA	[50]
			Au nanoparticles	Kapton	220 V, 1.8 mA	[21]
Nanoarrays	Surface structures	Template method, deposition, etching, printing	Water drops	Polyethylene (PE)	41 V, 1.3 μA	[94]
			Skin	PDMS/ZnO	87.5 V, 17.9 μA	[96]
			CNT	Boron nitride	17 V, 70 nA	[97]
Nanofiber surface and patterns	Surface structures made of nanofibers	Electrospinning	Polysuccinimide (PSI)	Perfluoroalkoxy alkane (PFA)	2 V, 2.5 μA	[102]

Table 2. Characteristics comparison of different processes for morphology preparation.

Processes		Advantages	Weaknesses	Materials	References
Chemical deposition		Thickness controllability, high reproducibility, low cost	1. Slow growth rate; 2. Deposition structures are disordered; 3. Chemical substances may have impacts on the environment	Metals, semiconductors, polymers	[96, 103]
Chemical etching	Wet etching	Processing rate controllability, high precision, low cost,	1. The solution is corrosive; 2. Etching depth is difficult to control;	Metals, semiconductors, ceramics, polymers	[48]
	Plasma etching	High precision, controllability, uniformity	High equipment costs, complex processes	Metals, semiconductors, glass, ceramics, polymers	[104]
	Reactive ion etching	High precision, controllability, uniformity	High equipment costs, complex processes	Metals, Semiconductors, glass, ceramics, polymers	[105]
BCP self-assembly		Orderly structure, high precision, low cost	1. Complex preparation conditions, high requirements for equipment and operation; 2. Organic solvents may have impacts on the environment.	Metals, polymers, silicon-based material	[49]
Template method		Structure controllability, high reproducibility, simple preparation, low cost	1. Precision is determined by the precision of the mold; 2. Difficulty in demolding	Metals, semiconductors, glass, ceramics, polymers	[38, 47, 106–110]
Imprinting		High precision, high repeatability, simple process	1. Mold preparation is complicated; 2. Mold size is limited.	Metals, semiconductors, glass, polymers	[51]
Crumpling method	Stretching	Simple process, controllable structures, wide applications	1. Small preparation size; 2. Low wrinkle regularity; 3. Low reproducibility; 4. Possible mechanical damage	Elastomers, films	[111]
	Heat treatment	Wide application, non-contact machining, simple process	1. Possible heating damage; 2. Low wrinkle regularity	Metals, glass, ceramics, polymers	[112]

(Continued.)

Table 2. (Continued.)

Processes		Advantages	Weaknesses	Materials	References
Breath figure method		Controllable pores, simple process, low cost	1. Disordered pore size; 2. Difficult to control the humidity of the preparation	Copolymers and polymer films	[113]
Physical etching	Photolithography	High precision, controllable, mature process, capable of multi-layer processing	1. High cost and operation technology requirement; 2. Complex process	Silicon, semiconductor materials, metals, glass, polymers	[114]
	Laser etching	High precision, non-contact machining	1. High equipment costs; 2. Possible high- temperature damage	Metals, semiconductor materials, ceramics, polymers	[115]
Printing	Inkjet printing	High precision, flexibility, non-contact machining	1. Slow printing speed; 2. Poor adhesion and durability	Paper, plastics, glass, metal, ceramic	[24, 116]
	Electrospinning	High precision, controllable fiber structure, low cost, wide applications	1. Fiber structure may be disordered; 2. Precise control of parameters is required.	Polymers, nanomaterials	[75, 117]
	Extrusion based printing/direct ink writing	Rapid preparation based on modeling, easy to handle, low cost, wide applications	1. Limited printing accuracy and material types; 2. Low structural strength of printed parts	Plastics, Elastomers	[79, 85, 118–120]
	Screen printing	Durability, low cost, wide applications, suitable for customized printing	1. Low precision; 2. Not suitable for large-area printing	Paper, plastics, glass, metals, wood	[121]
	Spray-coating	Efficiency, can be used on all shapes of surfaces, low cost	1. Difficult to control coating thickness; 2. Material surface must be pre-treated.	Metals, plastics, wood, glass, ceramics	[122]
Sputtering		Adhesion, controlled coating thickness	1. Low preparation efficiency; 2. High cost; 3. Need to cool down the substrate	Metals, semiconductors, ceramics, plastics	[123]

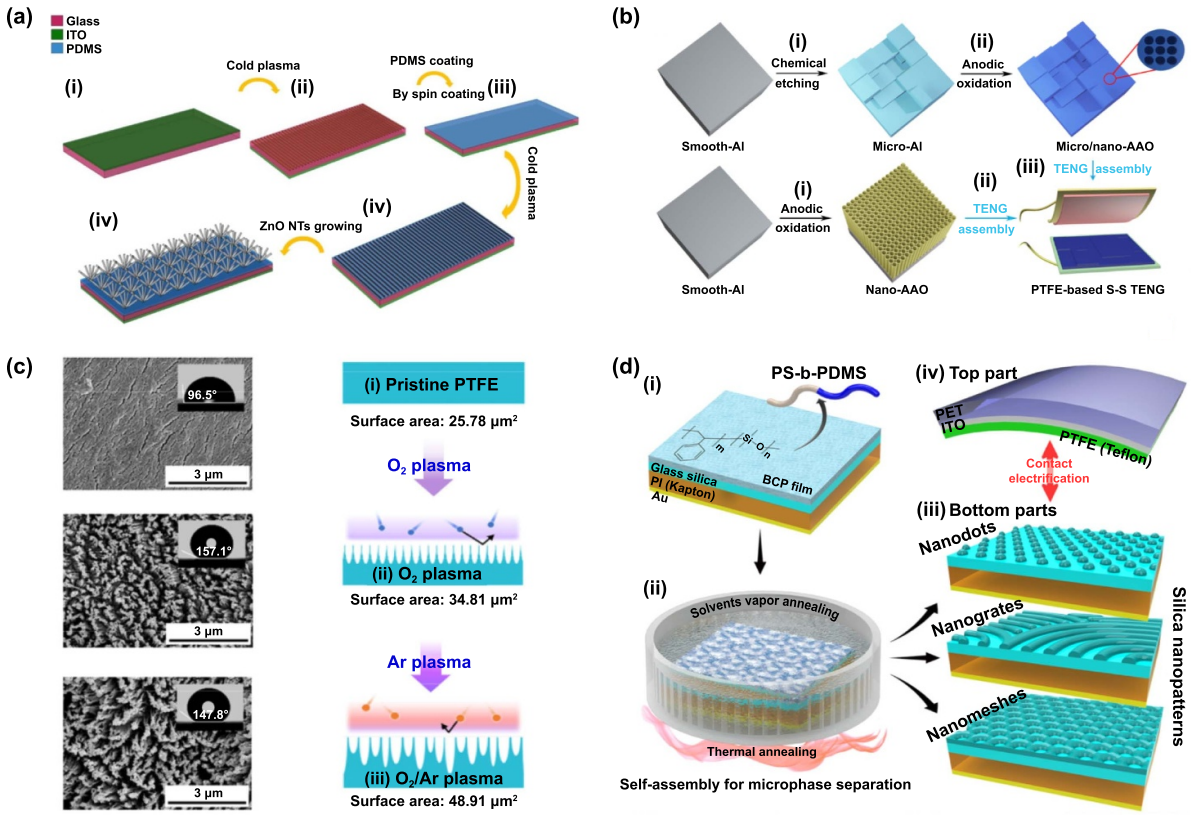


Figure 10. Chemical processes for micro-/nano-structures preparation. (a) Chemical deposition method for preparation of ZnO nanotubes. [96] John Wiley & Sons. © 2023 Elsevier Ltd. All rights reserved. (b) Wet chemical etching and electrolytic etching for preparation of micro-/nano-structures. Reproduced from [48] with permission from the Royal Society of Chemistry. (c) Reactive ion etching for preparation of PTFE micro-/nano-structures. Reproduced from [104], Copyright © 2021, Tsinghua University Press and Springer-Verlag GmbH Germany, part of SpringerNature. (d) BCP self-assembly process for preparation of nanostructures such as spheres, lateral cylinders, and hexagonally perforated lamellae. Reprinted with permission from [49]. Copyright (2014) American Chemical Society.

seed layer was formed on the PDMS surface by immersing it in a zinc acetate solution to provide primary ZnO particles. Then, the substrate was placed in a liquid-phase environment, and ZnO nanotubes were grown on its surface using an aqueous solution containing Zn^{2+} as a precursor. These ZnO nanotubes enhanced surface roughness and hydrophobicity, thereby augmenting output voltage and power. The single-electrode TENG with a size of 3 cm × 3 cm was capable of generating an open-circuit voltage of 87.5 V and a short-circuit current of 17.9 μA, with a maximum power of 36.16 $\mu W \cdot cm^{-2}$.

PU foams are highly flammable. Similarly, Phosphorus-doped mesoporous carbon (PMC) for modification could enhance the flame retardancy and triboelectric properties of PU foam [103]. The fabrication process involved sequentially immersing PU foams in positively charged chitosan solutions and negatively charged alginate-stabilized PMC aqueous dispersions, resulting in PMC-filled multilayers. This TENG exhibited an open-circuit voltage of 158 V and a short-circuit current density of 2.26 $\mu A \cdot cm^{-2}$, and demonstrated self-extinguishing properties when subjected to a butane flame.

The etching process can be used to prepare the surface morphology of the templates during the molding-release process, and can also be utilized directly to prepare the surface morphology of the friction layer. Common etching methods are physical etching methods (such as photolithography and laser etching, electrolytic etching), and chemical methods (such as wet chemical etching, plasma etching and reactive ion etching methods). For wet etching, chemical reagents are used to deal with the materials, so that part of the surface materials are dissolved. Wet etching has the advantages of high processing precision, low cost, wide applicability, etc, commonly used in the preparation of microporous and nanotube structures. Four distinct aluminum oxide microstructures were fabricated by wet etching and electrolytic etching [48], as shown in figure 10(b). Nanoporous anodic aluminum oxide (micro-/nano-AAO) films were fabricated using the anodic oxidation process twice, with aluminum and graphite flakes serving as the anode and cathode, respectively. The oxidation process involved immersion in oxalic acid under constant pressure, followed by immersion in the solution of chromic and phosphoric acid to remove the oxide film, resulting in nanoporous AAO films. Subsequently, samples were etched in 5 wt% phosphoric acid solution for less than 10 min to achieve AAO nanofilms with varying pore sizes. Micro-/nano-structured aluminum with a stepped microstructure was obtained by immersing

aluminum sheets in a solution of $CuCl_2$ and HCl for 8 min, followed by annealing at 200 °C for 1 h to enhance oxide microstructure formation. Micro-/nano AAO was obtained by anodizing microstructure aluminum in an oxalic acid solution for 1 h. TENGs were constructed by pairing each of these four alumina morphologies with a polytetrafluoroethylene (PTFE) film. The study demonstrated a positive correlation between contact area and measured charge density by adjusting the diameter of alumina nanotubes, thereby controlling the contact area of the friction layer.

Plasma etching uses plasma formed by excited gas to etch the surface of a material by ion bombardment. The process and morphology are shown in figure 10(c). Reactive components were often added to the gas to enhance the etching effect. This method offered precise control, high speed, and controllability, making it prevalent in microstructure fabrication. The inductively coupled plasma etching method, which was commonly used for PTFE-based TENG, was a method to fabricate microstructures on the surface of polymer materials using CF_4, O_2, and Ar. Due to the toxicity of CF_4, an alternative two-step method employing O_2 and Ar bipolar pulsed plasma etching, without CF_4, was proposed by Prada *et al* [104].

Correspondingly, reactive ion etching is also an etching method that utilizes ion-induced chemical reactions. It forms a damage layer on the surface, facilitating easy etching, and promotes chemical reactions for achieving anisotropic etching. The surface of the substrate to be processed is exposed to a plasma in which reactive ions react chemically with the material on the surface. Strictly speaking, reactive ion etching is a combination of physical and chemical actions. An environmentally resistant non-contact TENG was prepared by Han *et al* [105]. SiO_2 and PTFE particles were dispersed within thermoplastic PU and PDMS, respectively. A chamber pressure of 15 mTorr was employed for etching and fluorinating the PDMS-PTFE composites with (tridecafluoro-1,1,2,2-tetrahydrooctyl) trichlorosilane. This process aimed to enhance oil resistance and generate hydrophobic and oleophobic rough surfaces.

The template method is employed for morphology preparation and can be categorized into hard and soft template methods based on template characteristics. For modification of TENG topography, the hard template method typically involves solely physical changes, while the soft template method induces chemical alterations. Soft template structures are primarily upheld by weak intermolecular or intramolecular interactions, constituted of ordered aggregates from amphiphilic molecules like micelles, BCPs, vesicles, biomolecules, and liquid crystals. With a particular templating agent concentration, precursor growth is directed, leading to the formation of nanostructured materials with specific shapes. Deionized water was used as a soft template to prepare porous PTFE films for TENGs [106]. The porous PTFE films were obtained by mixing the hydrophobic PTFE solution with deionized water, and pouring the mixed suspension into a mold for curing and drying the film layer. Deionized water was used as a soft template, and different porosities were created by adjusting the volume ratio between deionized water and PTFE solution, and were removed using evaporation.

Poly(ethylene oxide)-b-poly(propylene oxide)-b-poly(ethylene oxide) triblock copolymer (P123) triblock copolymer micelles were utilized as mesoporous soft templates for self-powered NH_3 sensor [47]. P123 was first stirred to generate worm-like micelles, which were adsorbed on the graphene oxide (GO) surface through hydrogen bonding interactions. Then, heterostructures were prepared by *in situ* polymerization of pyrrole on the surface of P123 micelles-GO nanosheets.

Self-assembly is a method in which microscopic structural units naturally come together to create an organized structure. In the process of self-assembly, the basic structural units are spontaneously organized or aggregated into a stable structure with a certain regular geometrical appearance by interactions based on non-covalent bonds. BCP self-assembly is a 'bottom–up' chemical approach that enables precise control over the nanostructure of a material, offering advantages such as cost-effectiveness and high resolution when compared to physical etching methods. In 2014, BCP self-assembly was pioneered to modify the morphology of TENGs. The process of BCP self-assembly is shown in figure 10(d). Nanostructured silica films and PTFE were used as paired surfaces for the TENG friction layers. The patterned microstructures were formed through microscopic phase separation of covalently linked polymers. Polystyrene block PDMS (PS-b-PDMS) was selected for the construction of nanostructures on silica surfaces by applying a coating mixture solution onto polystyrene-modified silica glass substrates and induced by annealing. PS-b-PDMS could self-assemble into spatially confined PDMS microdomains in PS substrates, with the type of morphology determined mainly by the molecular weight and volume fraction of each polymer block. Upon selective removal of PS and oxidation of PDMS, silica nanodots, nanogrids, and nanonets could be formed [49, 124].

2.3.2. Physical preparation process. There are various physical modification methods applied to the preparation of morphologies, and distinct types of topography are usually prepared in different methodologies [38]. Certain preparation methods are specific to only one type or a small number of types of morphology, such as templating (soft/hard templating and sacrificial templating) and imprinting for the preparation of micro-/nano-array structures and porous structures, crumpling for the preparation of pleated structures, and respiratory mapping for the preparation of porous structures. The electrostatic spinning method is used to prepare fibrous structures on the surface. In addition, other preparation methods can be applied in the preparation of various morphologies, such as physical etching (photolithography and laser etching), printing (inkjet printing, direct ink writing, electrostatic spinning, screen printing, laser printing, 3D printing), and sputtering. In addition, multiple preparation methods are often combined in the preparation of surface topography, with a certain process as

an intermediate process, e.g. in the silicon template method for the preparation of micro-structured geometric arrays, the silicon template is first physically etched, followed by molding-demolding fabrication process.

The hard template method involves using materials with specific three-dimensional structures and easily controllable shapes as templates. The materials to be shaped are deposited into the holes or on the surface of the template using a certain method. Subsequently, the templates are removed to acquire the desired nanostructured material. According to the characteristics of the template, the primary methods for surface morphology preparation are the hard template method and the soft template method. Hard templates, typically derived from sturdy materials, exist in two forms: self-sacrificial templates, which actively partake in the reaction, and removable templates, which remain passive during the reaction. Removable templates can be eliminated either physically or chemically post-structure formation [107, 108]. Typical hard templates comprise silicon dioxide, metal oxides, and ice templates. To create PDMS with geometrical microarray surfaces, a silicon wafer served as a template. Photolithography was employed to generate the inverse pattern of the desired structure on its surface. The PDMS precursor was then cast into the mold, cured, and subsequently peeled off to yield a PDMS film featuring a regular geometric array structure [38]. Sugar and salt could also serve as templates for creating porous structures. Sugar was used as sacrificial particles for crafting porous films by Cho *et al* [109] Initially, CNTs and sugar were integrated into the PDMS precursor, followed by thorough stirring and curing. After curing, immersion of the films in deionized water facilitated the removal of sugar particles, yielding CNT/PDMS porous films. These films were then integrated with poly(ethylene oxide) (PEO) film to fabricate TENGs. The inclusion of CNTs amplified both the contact area and conductivity. Meanwhile, the porous structure further augmented the contact area and consequently increased the sensitivity of TENGs, as shown in figure 11(a). Apart from incorporating sugar or salt particles into the precursor solution, alternative methods involved casting the precursor solution onto a sugar or salt backbone. Sugar cubes as mold were immersed into a blend of GO and PDMS precursors [110]. Employing a vacuum-assisted infiltration technique, they infused the mixture into the cubes, ensuring uniform dispersion within the pores of sugar. After curing, the samples were immersed in water to dissolve completely the sugar to obtain three-dimensional microporous polymers.

Imprinting offers another method to introduce structures onto a surface. Imprinting involves transferring surface patterns by placing a sheet between upper and lower molds, and altering material thickness through pressure or high temperature to create micropatterns on the surface. The TENGs in vertical contact-separation mode based on imprinted PDMS films were fabricated. The surface morphology of PDMS film was prepared by imprinting, and nylon mesh was used as a template [51]. The process is shown in figure 11(b). The nylon mesh served as mold, the pattern embossed onto an aluminum foil using a press. Subsequently, the PDMS precursor solution was poured onto the aluminum foil mold and spread evenly, yielding a PDMS film featuring the surface morphology of the nylon mesh upon curing. Wrinkling serves as a prevalent physical technique for fabricating pleated structures. This method induces uneven deformation on surfaces by applying force or heat non-uniformly. A PDMS film with a higher buckling instability was stretched, and then the PDMS precursor solution was spun onto the film. After the upper layer of PDMS was cured, the force applied to the film was released. Due to the mismatch in the elastic moduli between the top and the bottom layers, the wrinkled surface was formed [111], as shown in figure 11(c). The breath figure method, utilized to fabricate thin films with orderly porous surfaces, involved modulating the microstructure of porous films by regulating parameters such as solvent type, concentration, humidity, and substrate properties. Initially, a solution containing a surfactant was deposited onto a substrate to form a thin film. The surfactant reduced the surface tension of liquid, leading to the formation of minute concave–convex structures on the surface of the film. Subsequent gradual solvent evaporation prompted surfactant molecules to aggregate, resulting in a film endowed with microscopic structures. As the solvent continued to evaporate, the surface tension of liquid film progressively increased, causing surfactant molecules to self-assemble into microstructures characteristic of the breath figure method. These microstructures continuously formed and dissolved on the liquid surface, resembling a breathing motion, hence earning the method its moniker, the 'breath figure method'. Once the solvent within the film completely evaporated, it left behind a porous structure. Plasma-enhanced breath figure was proposed to fabricate TENGs, combining the BF method and plasma technology for the first time [113]. The fabricated microporous fabric exhibited solvent tolerance, adjustable pore size, and excellent flexibility, which was suitable for large-scale manufacturing. Initially, nylon fabrics were submerged in a SEBS/chloroform solution to facilitate polymer solution attachment onto the fabric surface. Following controlled drying, a polymer coating featuring microporous structures was formed on the nylon fabric. Subsequently, plasma post-treatment facilitated the development of a solvent-insoluble cross-linking layer on the polymer surface, resulting in a surface microporous structure exhibiting outstanding solvent resistance.

Electrostatic spinning is a self-contained method for fabricating nanofibers, which is low-cost and widely applied. This technology can provide controllable fiber structures and high precision. The fundamental apparatus comprises three key components: a high-voltage power supply, a metal collector, and a spinneret. By adjusting parameters such as needle aperture and solution concentration, various fiber structures can be tailored to influence surface morphology. This technique is versatile, allowing for the creation of geometric arrays and structured fiber surfaces. For instance, large-area PVDF electrospun pyramid array films were produced by electrostatic spinning [75]. Utilizing a single-step self-assembly approach, they engineered these films for high-performance skin devices. Under an electrostatic field, individual fibers consistently passed through the apex, forming a suspended framework that delineated the angles of the pyramid.

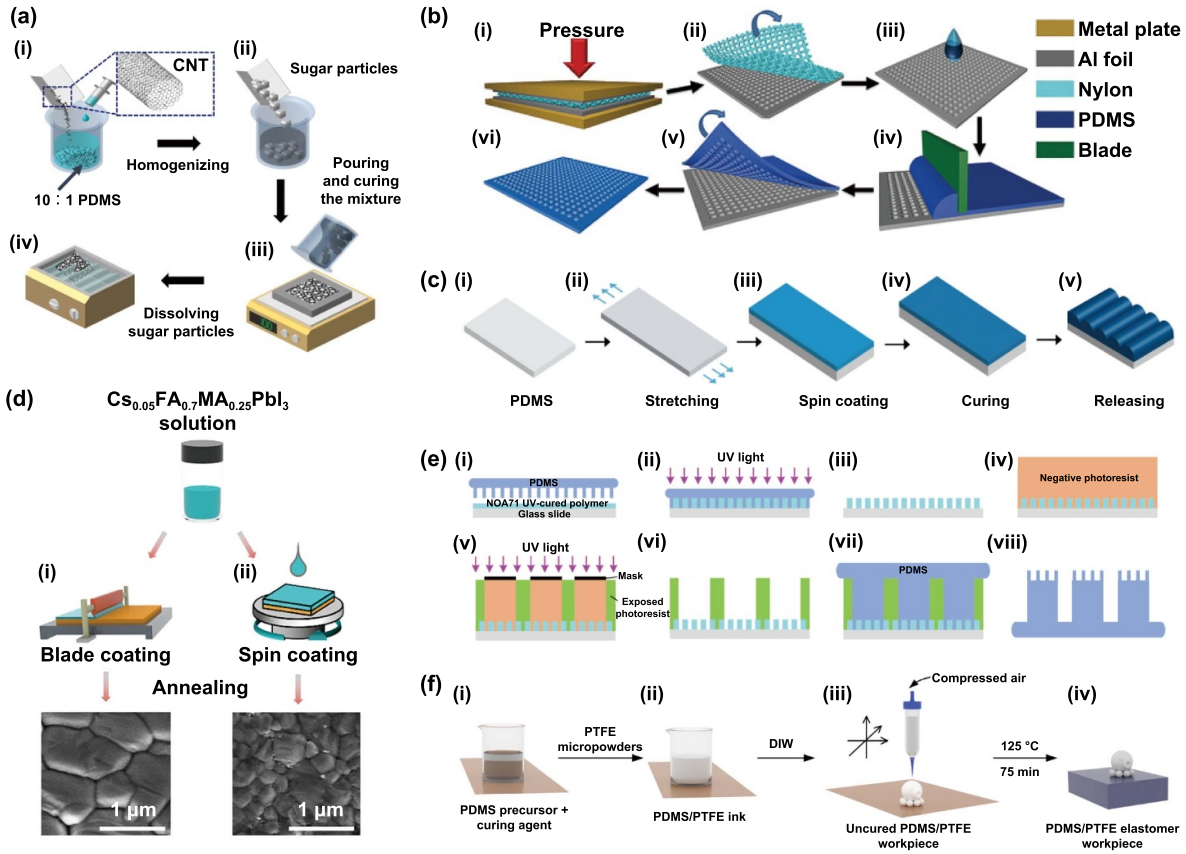


Figure 11. Physical preparation processes for TENG morphology. (a) Sugar template method for preparation of porous structures. Reprinted with permission from [109]. Copyright (2024) American Chemical Society. (b) Imprinting for preparation of nylon mesh structures. [51] John Wiley & Sons. © 2022 Wiley-VCH GmbH. (c) Mechanical stretching for preparation of wrinkled surfaces. [111] John Wiley & Sons. © 2022 Wiley-VCH GmbH. (d) Comparison of blade coating and spin coating for preparation of perovskite film. Reprinted from [117], © 2024 Elsevier B.V. All rights reserved. (e) Photolithography and micro-forming processes for preparation of graded microstructures. Reproduced from [114]. CC BY 4.0. (f) Direct ink writing for preparation of complex models. Reprinted with permission from [119]. Copyright (2020) American Chemical Society.

Through ongoing electrostatic spinning self-assembly, adjacent pyramids fused to form larger structures. Consequently, by regulating the self-assembly duration, control over pyramid size, ranging from microns to millimeters, was achieved. $Cs_{0.05}FA_{0.7}MA_{0.25}PbI_{0.3}$ perovskite films prepared by blade coating method and PVDF-graphene nanofibers prepared by electrospinning were applied to TENGs [117], as shown in figure 11(d). The smaller work function of the perovskite film and the larger effective contact area of the nanofiber layer improved the output performance of device. This 3 cm × 4 cm TENG was capable of generating an open-circuit voltage of 200 V and a short-circuit current of 16.3 μA, with a maximum transferred charge of 88.2 nC at 3 Hz and 50 N, and a power density of 11.32 $W{\cdot}m^{-2}$.

The primary physical etching techniques commonly employed for preparing TENG surface morphologies include photolithography and laser etching. Photolithography involves three core stages: exposure, development, and cleaning. Initially, the template is loaded into the photolithography machine, and masks are utilized to project a patterned beam of light onto the photoresist for exposure, inducing a chemical transformation. Subsequently, the template undergoes immersion in a developing solution to unveil the latent image within the photoresist. The chemically altered section of the photoresist dissolves, while the exposed region is etched, leaving the remainder unaffected. Ultimately, the template is submerged in a cleaning solution to eliminate any residual photoresist and contaminants. Photolithography and micromolding processes were used to fabricate PDMS layered structures [114], as shown in figure 11(e).

Laser etching is an etching method that uses laser light to process microstructures. Surface microstructures of silicone rubber could be generated by laser etching. The process could achieve continuous modulation of the TENG surface from hydrophobic to hydrophilic [115].

Inkjet printing is a non-contact ink deposition process that can be used to build free two-dimensional patterns on surfaces. The nozzles of printer are arranged in a tiny matrix, and each nozzle is usually controlled by a tiny electromagnetic device that can be turned on or off instantaneously. The number of nozzles and the density of their arrangement determine the resolution of the printer. Depending on the need of the printed

image, the printer opens the corresponding nozzle at a specific position to eject ink from an ink cartridge or capsule to produce the target surface pattern. A highly conductive MXene layer was inkjet-printed onto the hydrogel surface. Based on this method, wearable sensors with high sensitivity and wide detection range were prepared [116]. Inkjet printing was also utilized to prepare the surface morphology of one-dimensional stretchable fibers in wearable triboelectric sensors [24].

Extrusion-based printing (direct ink writing) is a 3D manufacturing technology, and extrusion-based printers are usually equipped with a controlled platform, suitable syringes and nozzles [79]. The ejection of ink from the nozzle is achieved through the application of pressure to the syringe. Control over ink flow is attained by regulating both the pressure and the dimensions of the printer nozzle. Acrylonitrile butadiene styrene/acrylate, oligomeric acrylate, and oligomeric PDMS elastomers stand out as the prevailing fillers utilized in TENGs [118]. This type of printing can be used to construct TENG surface patterns and also to prepare 3D skeletons for porous structural materials with controlled pore sizes. A composite ink was made by mixing PDMS and PTFE powder [119]. This innovative ink was tailored for extrusion-based printing (direct ink writing) applications, facilitating the 3D printing of elastomers, as shown in figure 11(f). Incorporating PTFE into the elastomer not only elevated the viscosity of the precursor solution, rendering it compatible with extrusion-based printing but also heightened the electron affinity of the composite elastomer, thereby enhancing the output performance of TENGs. The exceptional compatibility between PTFE micro-powder and the nonpolar PDMS precursor solution ensured the ink possesses desirable rheological properties and prevented substrate hardening. A hierarchical porous structure TENG based on 3D printing was prepared by Chen *et al* [120]. Salt particles with diameters of 37 μm–75 μm and CNTs were doped into a composite ink, and an insole consisting of fibers was printed by extrusion. After curing, the salt was sequentially removed by immersion in water/ethanol solution and the CNTs were removed by immersion in lipase solution. A layered porous structure was obtained.

Furthermore, based on 3D printing, 4D printing technique has been proposed to be applied to TENG morphology preparation using shape memory materials. Huang *et al* prepared TENG from SMP using extrusion-based printing, capable of generating a power density of 56 $mW{\cdot}m^{-2}$ [85]. The PU/PLA composite was heated to 210 °C in a fused deposition molding printer and extruded through small openings to form SMP filaments, which were printed as 3D microstructure surfaces based on design modeling. Electrodes were prepared with silver nanowires using spraying. In addition, this TENG prepared by 4D printing had good damage resistance and could achieve morphological recovery under heated conditions. When the friction layer collapsed, the device was placed in an oven at 60 °C for 30 s, and the morphology could be restored to the initial state.

Screen printing, a technique for imprinting two-dimensional patterns, involves the passage of ink through a screen onto various substrates like fabric, acrylic board, or metal plate. This method relies on utilizing a screen to reproduce the desired pattern. Initially, a suitable screen material is chosen, and a screen plate is prepared. The plate is coated with photographic emulsion, and the pattern is transferred onto it via exposure, leaving the unexposed areas as openings for the pattern. The resulting stencil is affixed to the screen printer, ensuring a snug fit with the printing material. Ink is applied onto one end of the stencil and spread across its surface using a squeegee. The pressure and speed of the squeegee regulate the ink flow through the openings of stencil onto the material, forming the intended pattern. The screen-printing technique was also capable of TENG manufacturing. Nylon ink, Ag ink, and additional layers of nylon ink were sequentially screen-printed and dried onto a textile substrate. This process yielded a dense three-layer film structure, comprising a monolithic silver ink layer sandwiched between layers of dense nylon ink. This structure was further combined with PTFE to create a monolithic-grade TENG [121].

Spray-coating, an extensively used method in surface coating, facilitates the deposition of materials onto surfaces. This technique typically involves specialized equipment including a spray gun, a compressed air source, and a paint supply system. The spray gun employs compressed air to apply the coating onto the surface, directing the paint through the nozzle at high velocity. Key parameters such as nozzle size, spray angle, and spray pressure are finely tuned to ensure uniform surface coverage. A TENG was developed using a spray-coating process with a filipin protein patch film [122]. Initially, protein solution was prepared by dissolving filipin protein powder in deionized water. This solution was then loaded into a spray gun and applied multiple times onto the substrate surface before drying. The morphology of the resulting silk-fibroin patch film could be adjusted by varying the spraying duration. Exploiting the electron-loss properties and textured surface of silk protein, the fabricated TENG exhibited remarkable performance, achieving a maximum voltage output of 213.9 V and a high power density of 68.0 $mW{\cdot}m^{-2}$. Sputtering is also a physical deposition method used for surface morphology construction. The substrate material and the target are placed in a vacuum chamber. The sputtering target is excited using different excitation sources, such as electron or ion beams. Atoms or molecules on the surface of the sputtering target form a plasma that is ejected at high speed onto the substrate surface. The atoms or molecules deposited onto the substrate surface diffuse across the surface and are gradually deposited, eventually forming a homogeneous thin film. Sputtering can be used to build geometrical micro-/nano-arrays and can also be used to construct folded structures. For example, the sputtering process could be used for preparing waterproof dual-wavelength layered hierarchical wrinkled structures on artificial friction electric skin [123]. The surface of SEBS was first plasma-treated with a linear ion source to form micro-wrinkles. PTFE and CNTs were mixed to make a composite target. The composite target was then coated on the SEBS surface by sputtering to form secondary nanostructures.

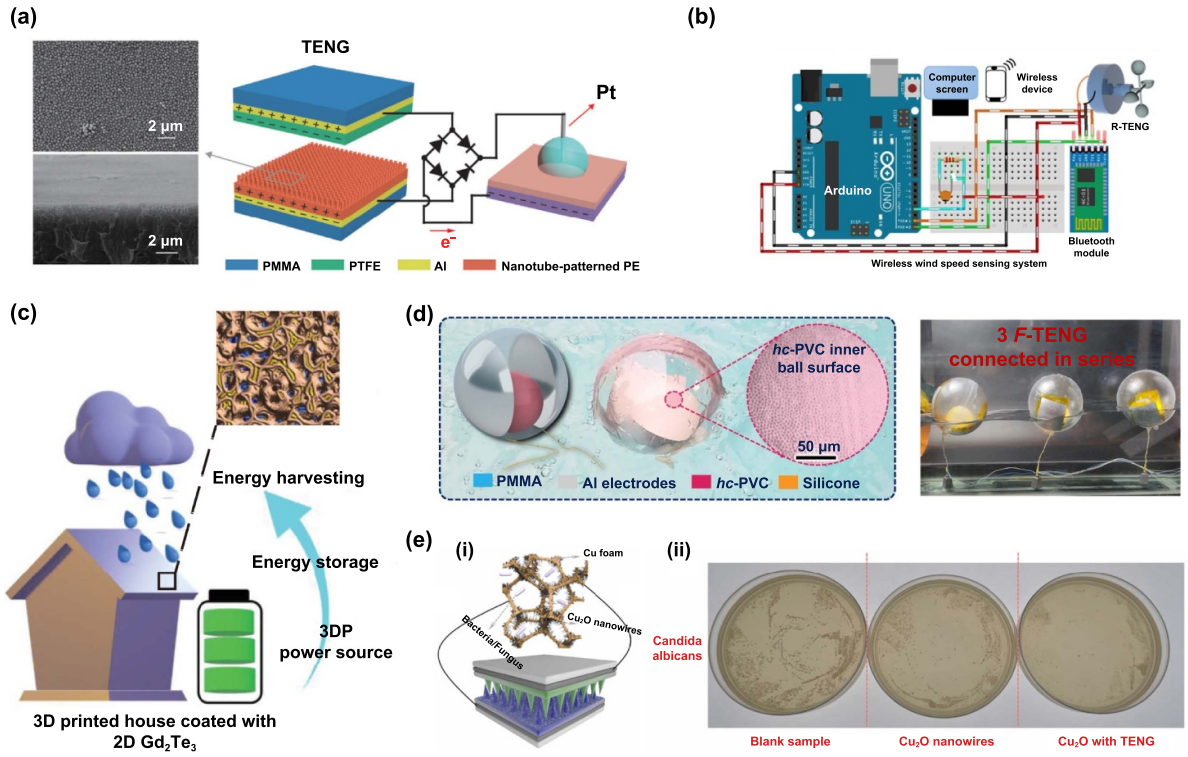


Figure 12. Applications of TENGs micro-/nano-morphology in energy harvesting. (a) TENG modified with surface nanotubes to power an electrowetting display system. [53] John Wiley & Sons. © 2024 Wiley-VCH GmbH. (b) TENG modified by electrochemical anodizing to power small electronic devices. Reprinted with permission from [126]. Copyright (2023) American Chemical Society. (c) TENG with porous structures used for harvesting raindrop energy. [54] John Wiley & Sons. © 2022 Wiley-VCH GmbH. (d) TENG with honeycomb porous structure for ocean energy harvesting. Reproduced from [52]. CC BY 4.0. (e) TENG with microvertebrate array structures for powering Cu_2O nanowires to bolster sterilization efficacy. Reprinted from [55], © 2022 Published by Elsevier Ltd.

3. Application of TENG with special morphology

3.1. Energy harvesting

TENGs with morphology modification can be utilized in energy harvesting and wearable sensors. Micro-/nano-structures can effectively enhance the amount of charge transferred in the triboelectric process. Therefore, in the process of energy harvesting, micro-/nano- structures play a role in improving the output performance [125]. When operated as self-supplied energy sensors, the morphologies play a role in improving the sensitivity of the device.

On the one hand, TENGs with special morphologies can be used as power generation devices to collect mechanical energy from natural phenomena or human activities. Then TENGs convert collected energy into electricity for energy supply to other devices. For example, a contact separation TENG based on PTFE film and PE film modified with nanotubes powered an electrowetting display (EWD) system [53]. The system is shown in figure 12(a). The nanotube pattern on the PE surface could increase the contact area, which helps to boost output performance. The 4 cm $\times$ 4 cm TENG generated an open-circuit voltage of 160 V and a short-circuit current of 32 μA. This high performance enabled the EWD to exhibit excellent responsiveness when powered by the TENG. The EWD could rapidly switch colors within 25 s and revert to its initial state within 1.2 s. This enabled the TENG to dynamically switch colors. The EWD could quickly switch colors within 25 s and return to its original state within 1.2 s, enabling TENG to trigger dynamic color displays. The electrochemical etching process to enhance the surface area of rotational TENG was performed [126], as shown in figure 12(b). The etched copper film surface was distributed with irregular nanopatterns of more than 100 nm. The TENG with microstructure copper film produced electrical outputs of about 17 V and 0.6 μA, under the force of 7 N and frequency of 5 Hz. In contrast, The TENG with the area of 2 cm $\times$ 2 cm based on smooth copper films produced about 10 V and 0.4 μA under the same conditions. It demonstrated that surface pattern was effective in output enhancement. By affixing four TENGs to wind cups, with positive and negative layers interconnected to harness rotational energy, such as wind power, these systems could effectively power small electronic devices. Porous TENGs coated with gadolinium telluride (Gd_2Te_3) were laid on the roof to collect energy from raindrops, as is shown in figure 12(c) [54]. A surface hydrophobicity modulation method was achieved by laser etching on a silicone rubber surface [115]. The micro-wall-like structures could be fabricated on the silicone films by laser etching. By controlling the laser power and scanning speed, the roughness of the silicone rubber surface could be regulated, thereby changing the hydrophobicity and the contact area of the surface. The

contact angle of water droplets on the TENG surface was up to 158°. The TENG based on superhydrophobic surface silicone rubber could operate continuously and stably in high humidity environments, enabling energy harvesting from raindrops outdoors. In a simulated rainfall with a flow rate of 10 ml·s^{-1} and a height of about 20 cm, the device achieved a maximum output voltage of 30 V and a maximum output current of 14 μA. A magnetically assisted noncontact TENG for wind and blue energy harvesting was prepared [127]. The TENG consisted of a magnetically responsive PDMS/FeCoNi layer and a non-magnetically responsive Al/Ni fabric layer, in which a PDMS film with 1.5 μm was prepared by blending a PDMS precursor with magnetic Fe–Co–Ni powders and using DVD disc as mold. Through the integration of magnetic responsiveness with TENG technology, a noncontact TENG was developed. This design not only enabled easier encapsulation of the device but also mitigated the risk of damage, thereby promoting extended device lifespan. Honeycomb porous structures on the polyvinyl chloride waste were fabricated for ocean energy harvesting [52], as shown in figure 12(d). The micro-pores were regularly distributed between the layers, and the porous array facilitated the enhancement of the contact area, hydrophobicity, and concentration of electrons. Compared to the TENG with flat films, the TENG with honeycomb structure presented an enhancement of 2.7 times in voltage and 4 times in power density. A flower-like TENG was fabricated to monitor wind speed and direction [56]. This TENG consisted of a flower-like frame structure and an array of TENG units. Each unit had a polyimide film and a porous aluminum film as a friction layer. When the wind blew, the TENG generated an electrical signal through a contact-detachment process, which not only effectively collected wind energy, but also served as a self-powered wind sensor to detect wind speed and direction. At a wind speed of 15 m·s^{-1}, each bionic TENG unit had an open-circuit voltage of 61.7 V and a short-circuit current of 0.34 μA. Microvertebrate array was prepared on a PU layer using maskless direct image lithography and subsequently chemically modified the surface through fluorination [55], as shown in figure 12(e). The fluorination treatment induced microbumps, microscratches, and introduced fluorine elements on the microcone surface, thereby amplifying the surface contact area and enhancing electron affinity. The resulting TENG, based on the fluorinated PU layer, achieved an output current of 22 μA, a five-fold increase compared to a flat TENG. This specialized TENG was found application in biological settings for mechanical energy harvesting and powering cuprous oxide (Cu_2O) nanowires to bolster sterilization efficacy against Gram-negative *E. coli* and Gram-positive *S. aureus*.

3.2. Self-powered sensors

On the other hand, TENGs with morphological modifications serve as self-powered sensors for real-time monitoring of human activities and work environments. The morphology of the surface and internal structure serves to enhance the sensitivity of the device to better match the working environment. The topographically modified TENGs can convert mechanical energy generated from human physiological activities such as walking and breathing into electrical signals for real-time monitoring of human health. The smart insole capable of monitoring foot pressure and health was made up of layered porous TENG array [45], as shown in figure 13(a). The smart insole could analyze movements and assesse postures for sports such as tai chi and throwing based on the forces applied. In addition, the smart insole could be utilized to analyze healthy information such as the body weight and foot force status of people, thus enabling foot health diagnosis. An electronic skin with micro-pyramid array structures was based on wet heterostructured electrified jets. Based on the triboelectric effect, the electronic skin was ultra-thin, ultralight and breathable, and had good performance in daytime radiation cooling, pressure sensing and bioenergy harvesting, etc [75, 128], as shown in figure 13(b). The hydrogel TENG decorated with micro-pyramid array was capable of detecting subtle body movements, including breathing, knuckle movements, droplet movement, and airflow [129], as shown in figure 13(c).

What's more, the modified shape of TENGs lends itself to signal detection and monitoring applications in industrial and machinery sectors. Ferroelectric materials and surface modification played a role in increasing the surface charge, which enhanced the output performance of TENG [130]. Fibrous mats were fabricated by electrostatic spinning and electrospraying using poly(vinylidene fluoride-trifluoroethylene) copolymer and polyamide 11 (nylon-11), and poly(3,4-ethylene dioxythiophene): poly(styrenesulfonate) nanofillers were added to the composite solution to improve the ferroelectricity properties. This TENG exhibited sensitivity to variations in wind speed, with the output voltage escalating from 3 V to 8.8 V as the airflow velocity escalates from 2 m·s^{-1}–5 m·s^{-1}. The fiber-based TENG served as an airflow sensing component in heating, ventilation, and air-conditioning systems, potentially contributing to energy conservation efforts, as shown in figure 13(d). Furthermore, a TENG for road safety monitoring had linear dot patterns on the surface. The aluminum electrode modified by zinc sulfide (ZnS) nanostructures was used as the positive friction layer, with the modification achieved by a modified hydrothermal method. The overhead projector transparencies with different line patterns directly printed on them were used as the negative friction layer [58]. With a 5 cm × 5 cm area, this TENG generated an open-circuit voltage of 420 V and a short-circuit current of 83.33 mA·m^{-2}. TENGs with cross-parallel dots were utilized for road safety monitoring, as is shown in figure 13(e). The surface microstructure enabled the device to increase the open-circuit output voltage by about 11 times and the short-circuit current density by about 17 times compared to TENG with smooth surface.

4. Summary and outlook

In this review, we present common TENG morphologies. These morphologies primarily bolster output performance by augmenting surface roughness, consequently expanding the

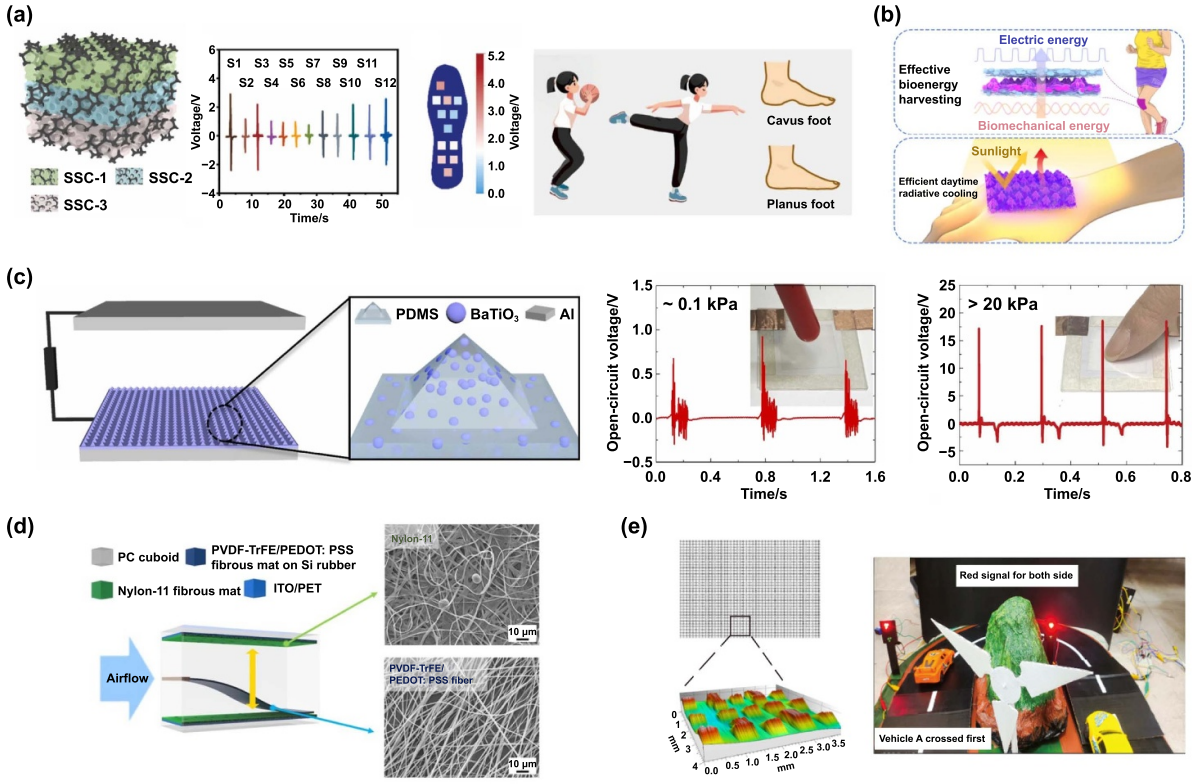


Figure 13. Applications of TENGs micro-/nano-morphology in self-powered sensors. (a) Layered porous TENG array for a smart insole to monitor foot pressure and health. Reprinted from [45], © 2024 Elsevier Ltd. All rights reserved. (b) Electronic skin with micro-pyramid array structures. Reproduced from [75]. CC BY 4.0. (c) TENG decorated with micro-pyramid array capable of detecting subtle wind and body movements. Reprinted from [129], © 2024 Elsevier Ltd. All rights reserved. (d) Fiber-based TENG for airflow sensing. Reproduced from [130]. CC BY 4.0. (e) TENG with cross-parallel dots utilized for road safety monitoring. Reprinted with permission from [58]. Copyright (2023) American Chemical Society.

contact area. The common morphologies consist of surface-only and surface-interior-distributed structural units. Porous structure is a representative morphology distributed on the surface and inside, while array structures, wrinkled structures and fiber structures are surface morphologies. Among the array structures, pyramidal and porous structures are the more common morphologies that play a significant role in performance enhancement. Pyramid arrays are good for increasing the surface area. For porous structures, the surface pores enhance the contact area and the pores inside the film improve the compressibility. The fiber structure can also play a good role in modification. For fiber-based TENG, the fiber structure and the presence of pores between the fibers enlarge the specific surface area [74]. On the basis of primary structures, secondary structures can be prepared to fulfill functions that enhance the primary structures, such as increasing sensitivity or maintaining contact angle. Such structures are called hierarchical structures. Geometric arrays, wrinkled structures, fiber structures and other structural patterns can be designed as hierarchical structures [112]. We further divide the methods for preparing multiple morphologies into physical and chemical methods, and describe them separately.

In addition, performance enhancement of TENG shape modification should be comprehensively investigated in the following aspects (as is shown in figure 14):

First of all, the working mechanism of TENG should be deeply explored and the perfect working model of TENG should be established [131]. Triboelectrification is a common natural phenomenon. According to current research, the greater the variance in electron affinity between two materials is, the higher the resultant surface charge density. Enhancing the microstructural surface can increase the contact area, thereby improving output performance. However, the precise micromechanics governing the charge transfer process remain elusive. Therefore, it is necessary to continuously explore the relationship of quantitative matching between micro-/nano-structures and charge generation. In addition, the problem of matching friction materials, optimal morphology and friction charge should also be theoretically solved.

Secondly, the preparation process should be simplified to enhance environmental friendliness. Currently, to obtain high performance, some TENGs are prepared with high equipment conditions or by toxic reagents. When these TENGs are used to harvest natural energy, they may cause environmental pollution. The selection of biodegradable materials and reduced use of toxic reagents can enhance environmental friendliness. In addition, simplifying the morphology preparation process can reduce the cost and improve the applicability of TENGs. Simplifying the preparation process and

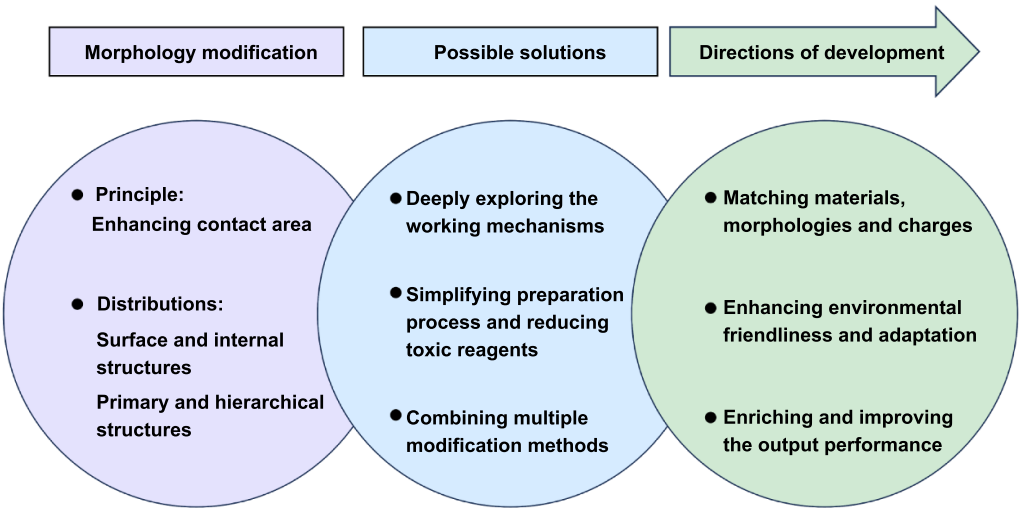


Figure 14. The perspective of TENGs with surface and internal morphologies.

utilizing natural materials or non-toxic reagents aligns more closely with the principles of eco-friendly energy production [16, 132].

Thirdly, when preparing morphology, multiple modification methods should be combined to enrich the device performance. The modification method can improve the performance of the triboelectric materials, so as to enhance the environmental adaptation of the device. TENGs are widely used in many areas, for which various structures have been designed, such as disc, spiral and sphere, etc. The output performance of TENGs is affected by many environmental factors, such as temperature and humidity. Therefore, the development of environmentally-resistant materials for friction layer is effective in improving the environmental adaptation. There are two kinds of material modification: physical modification and chemical modification. Physical modification is to modify the surface morphology of the material, and chemical modification refers to the introduction of special functionality to enhance the existing properties (such as improving mechanical properties) or special properties (such as hydrophobicity, self-repairing properties and self-cleaning properties). Moreover, integrating layered structures into the microscopic surface architecture not only amplifies effective contact area but also significantly enhances the longevity, signal consistency, and resilience of TENGs. Functionalized TENGs exhibit superior environmental adaptability and expand their applicability across diverse domains.

In addition, the TENGs developed presently exhibit characteristics of 'high voltage, low current'. The voltage can reach hundreds of volts, while the current can only be at the level of microamps, thus only producing a low output power. In addition, humidity also has a more pronounced effect on TENG performance, so it is also important to improve output stability. This requires researchers to further improve the output performance of the devices through material, morphology and structural design.

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